
Quantifying Cross-Domain Knowledge Distillation in the Presence of Domain Shift

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Abstract

This paper presents a theoretical investigation into the generalization capabilities of cross-domain knowledge distillation. Utilizing a high-dimensional asymptotic analysis of a linear teacher-student model, we characterize the excess risk while accounting for both model and covariate shifts. Our results provide a formal guarantee for the efficacy of distillation: even when the source and target domains differ substantially, there still may exist a regime where the student model achieves superior generalization ability over the student-only baseline. Moreover, we identify a *crossed double descent* phenomenon: the excess risk can vary non-monotonically with the teacher’s and student’s dimension-to-sample-size ratios. These results provide rigorous insight into when and why distillation helps across domains.

1. Introduction

The success of modern machine learning tasks typically requires the availability of large-scale labeled datasets. However, collecting labeled data for a new target task is often challenging and expensive. When data in the target domain is scarce, it is possible to leverage labeled data from related source domains. Knowledge distillation (KD) (Hinton et al., 2015), originally proposed for model compression, is a popular technique that transfers knowledge from a capable teacher model trained on a source domain to a smaller student model. This is achieved by guiding the student model to mimic the teacher model’s outputs. The extra informa-

tion in the teacher’s predictions often improves the student model’s performance when target domain data is limited. KD has recently achieved remarkable success across several fields including image classification (Radford et al., 2021; Li et al., 2024), speech recognition (Mingote et al., 2020), and language models (Gu et al., 2024; Agarwal et al., 2024).

We denote the source domain data and target domain data as $(\mathbf{X}_1, \mathbf{y}_1)$ and $(\mathbf{X}_2, \mathbf{y}_2)$, respectively. This work focuses on the following cross-domain KD process: a teacher model is first trained on the source domain data, and its predicted labels for the target domain inputs are then used to supervise the training of the student model by minimizing the per-sample objective function,

$$\mathcal{L}(\xi) = \xi \ell(y_2^t, y_2^s) + (1 - \xi) \ell(y_2, y_2^s), \quad (1)$$

where ℓ denotes the loss function, y_2 is the ground-truth label, y_2^t is the teacher’s predicted label, and y_2^s denotes the student’s prediction. The weight parameter ξ , known as the imitation parameter (Lopez-Paz et al., 2015), balances the contributions of the teacher’s predictions and the observed labels during training.

Cross-domain KD often suffers from a *shift* between the source and target domains. For instance, the source domain may consist of standard American English speech, while a region-specific voice assistant must handle local dialects. Another example is a face detection model trained on images of light-skinned individuals (source domain) being applied to images of dark-skinned individuals (target domain). Learning a discriminative predictor under such domain shifts between source and target domains is known as domain adaptation (Ganin et al., 2016). While much of the literature on domain adaptation has focused on improving the performance of KD, relatively little is understood about when – and how effectively – the student model can learn from the teacher in the presence of domain shift.

Recently, (Emrullah Ildiz et al., 2025) analyzed the weak-to-strong (W2S) generalization of linear models in a cross-domain setting, and identified the form of the optimal surrogate model. However, their analysis relies on the condition that the covariance matrices of the source and target domains are jointly diagonalizable, which limits its ability to capture the influence of eigenvectors. Moreover, their re-

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sults are restricted to the setting $\xi = 1$ (i.e., pure teacher supervision), leaving the trade-off between distillation and learning from observed student data unexplored. Motivated by these limitations, we take a step toward a more complete understanding of the performance gains of cross-domain KD for linear regression.

In this paper, we present a theoretical analysis of cross-domain KD in the context of linear models, leveraging tools from random matrix theory. For ridge regression, we study two cases: (i) a deterministic-parameter setting, in which the teacher and student parameter vectors are non-random; and (ii) a random-parameter setting, in which a shared parameter vector is drawn from a prior distribution. We also extend our analysis to the ridgeless regression setting. All proofs of the theoretical results are provided in the appendix. We summarize our contributions as follows:

- **High-dimensional risk characterization.** We derive precise high-dimensional asymptotics for the risk of cross-domain KD via a bias–variance decomposition. Our results reveal how the excess risk depends on the parameter vectors and the input distributions in both domains, generalizing the student-only setting of (Hastie et al., 2022).
- **Utility of cross-domain KD.** (“*Stones from other mountains can polish jade*”). Intuitively, large domain shifts between the teacher’s and student’s training data might limit – or even harm – the value of teacher supervision for the student’s generalization. Surprisingly, our analysis shows that even under substantial domain discrepancies, it is still possible to find an $\xi \in \mathbb{R}$ such that the student model can outperform the student-only baseline. The existence of such ξ depends on the geometry of the models and the covariance matrices of both domains.
- **Double descent phenomenon.** We demonstrate that the asymptotic excess risk in KD exhibits a non-monotonic behavior with respect to the dimension-to-sample-size ratios of both the teacher and the student. Moreover, by the theoretical results we identify and characterize a novel crossed double descent phenomenon in KD (see Figure 2). While prior studies have documented double descent in student-only settings (Hastie et al., 2022; Nakkiran et al., 2021), or focused on the specific case of $\xi = 1$ under isotropic conditions (Moniri & Hassani, 2025), this *crossed* behavior highlights an interaction between the teacher and the student that has not been previously characterized.

1.1. Related Works

Theory of KD. In recent years, a growing body of work has sought to understand the effects of KD. The theoret-

ical understanding of distillation began with (Phuong & Lampert, 2019), who initially investigated linear student networks. (Wei et al., 2021; Borup & Andersen, 2021; Das & Sanghavi, 2023; Pareek et al., 2024; Jeong & Chung, 2025) theoretically studied self-distillation, a variant of KD in which the student model has the same architecture as the teacher and is trained on the same data. (Menon et al., 2021) showed that a “Bayes teacher” providing true class probabilities can reduce the variance of the student’s objective, leading to improved performance. (Harutyunyan et al., 2023) proposed a framework that highlighted a delicate interplay among the teacher’s accuracy, the student’s margin with respect to the teacher predictions, and the complexity of the teacher predictions. From an information-theoretic perspective, (Dissanayake et al., 2025) quantified and explained the transferred knowledge and knowledge left to distill for a downstream task.

Cross-domain KD and domain adaptation. Many studies have explored various methods to address the domain shift problem in the field of KD. Empirical works include (Su & Maji, 2016; Kundu et al., 2019; Asami et al., 2017; Li et al., 2023; Xu et al., 2024; Tang et al., 2025). (Ye et al., 2024) proposed the Maximum Conditional Mutual Information method, which enables the teacher model to capture more contextual information to generate more accurate estimates of the Bayes conditional probability distribution. The emergence of large language models (LLMs) has brought new advancements, such as distillation across vastly different architectures and scalable cross-domain transfer. For more details, readers may refer to (Fedus et al., 2022; Ouyang et al., 2022; Yang et al., 2024). From a theoretical perspective, (Emrullah Ildiz et al., 2025) focused on the setting where the student is trained using only the teacher’s predictions, and analyzed the conditions under which the student can outperform the teacher in cross-domain KD.

Weak-to-strong generalization. Weak-to-strong (W2S) generalization (Burns et al., 2024), which concerns using predictions generated by a weaker teacher model to train a more powerful student model, is closely connected to KD. (Emrullah Ildiz et al., 2025) provided an analysis of ridgeless regression and proved that when using a weak model as the surrogate (teacher), W2S training can provably outperform training with true labels. (Charikar et al., 2024) assumed that the models are selected over a convex set, and quantified the gain of the weak-label trained strong model over the weak model. (Wu & Sahai, 2025) explored W2S generalization for classification in a spiked covariance model. (Medvedev et al., 2025) explained how W2S generalization can arise in random feature models described by two-layer networks. Theoretical research in this area has continued to grow, see (Dong et al., 2025; Shin et al., 2025; Moniri & Hassani, 2025; Oh et al., 2025), for example.

1.2. Notations

We use $\|\cdot\|$ to denote the spectral norm for matrices and the Euclidean norm for vectors, and $\|\cdot\|_F$ for the Frobenius norm of a matrix. Standard big-O and small-o notations are employed. Moreover, we denote $x_n = o_{a.s.}(a_n)$, if $x_n/a_n \rightarrow 0$ almost surely. For any sequences $a_n \geq 0$ and $b_n \geq 0$, we write $a_n \lesssim b_n$ if $a_n = O(b_n)$, and $a_n \sim b_n$ if both $a_n \lesssim b_n$ and $b_n \lesssim a_n$. We use $\delta(\cdot)$ to denote the indicator function, which takes the value 1 if the condition $\cdot$ holds, and 0 otherwise. Throughout the paper, c and C denote constants that may vary from line to line. For a random variable x , we use $x \sim D$ to indicate that x follows the distribution D .

2. Preliminaries

2.1. Problem setup

Suppose there are N_1 covariates $\{\mathbf{x}_j^{(1)}\}_{j=1}^{N_1}$ drawn i.i.d. from an M -dimensional source distribution D_1 and N_2 covariates $\{\mathbf{x}_j^{(2)}\}_{j=1}^{N_2}$ drawn i.i.d. from an M -dimensional target distribution D_2 . We consider a linear regression task specified by an unknown parameter vector $\beta_i \in \mathbb{R}^M$:

$$y_j^{(i)} = \beta_i^\top \mathbf{x}_j^{(i)} + \varepsilon_j^{(i)}, \quad i = 1, 2, \quad 1 \leq j \leq N_i,$$

where $\varepsilon_j^{(i)} \in \mathbb{R}$ is a zero-mean random noise term with variance σ^2 . For $i = 1, 2$ and $z \in \mathbb{C} \setminus \mathbb{R}^+$, define

$$\mathbf{X}_i = (\mathbf{x}_1^{(i)}, \dots, \mathbf{x}_{N_i}^{(i)}) \in \mathbb{R}^{M \times N_i}, \mathbf{y}_i = (y_1^{(i)}, \dots, y_{N_i}^{(i)})^\top,$$

$$\mathbf{Q}_i(z) = \left(\frac{1}{N_i} \mathbf{X}_i \mathbf{X}_i^\top - z \mathbf{I}_M \right)^{-1}, \quad \varepsilon_i = (\varepsilon_1^{(i)}, \dots, \varepsilon_{N_i}^{(i)})^\top.$$

We refer to the case where $D_1 \neq D_2$ as a *covariate shift*, and the case where $\beta_1 \neq \beta_2$ as a *model shift*.

Teacher Model: The teacher model is finetuned on $\{(\mathbf{x}_j^{(1)}, y_j^{(1)})\}_{j=1}^{N_1}$:

$$\begin{aligned} \beta_t &= \arg \min_{\beta} \left(\frac{1}{N_1} \|\mathbf{y}_1 - \mathbf{X}_1^\top \beta\|^2 + \lambda_t \|\beta\|^2 \right) \\ &= \frac{1}{N_1} \mathbf{Q}_1(-\lambda_t) \mathbf{X}_1 \mathbf{y}_1, \end{aligned} \quad (2)$$

where $\lambda_t > 0$ is the teacher regularization parameter. The risk of β_t when $M \sim N_1$ in the high-dimensional setting has been studied extensively in the literature such as (Do-briban & Wager, 2018; Hastie et al., 2022).

Student Model Trained with Cross-Domain KD: We use the pre-trained teacher model together with covariates $\{\mathbf{x}_j^{(2)}\}_{j=1}^{N_2}$ to generate predictions:

$$\mathbf{y}_2^t = (y_1^t, \dots, y_{N_2}^t)^\top = (\mathbf{x}_1^{(2)}, \dots, \mathbf{x}_{N_2}^{(2)})^\top \beta_t.$$

The student model is finetuned on the target domain data $\{(\mathbf{x}_j^{(2)}, y_j^{(2)})\}_{j=1}^{N_2}$ and the teacher's predictions $\{(\mathbf{x}_j^{(2)}, y_j^t)\}_{j=1}^{N_2}$, using the per-sample objective function defined in (1) with an imitation parameter ξ , as follows:

$$\begin{aligned} \beta_s &= \arg \min_{\beta} \mathcal{L}(\xi) = \arg \min_{\beta} \xi \left(\frac{1}{N_2} \|\mathbf{y}_2^t - \mathbf{X}_2^\top \beta\|^2 \right) \\ &\quad + (1 - \xi) \left(\frac{1}{N_2} \|\mathbf{y}_2 - \mathbf{X}_2^\top \beta\|^2 \right) + \lambda_s \|\beta\|^2 \\ &= (\mathbf{X}_2 \mathbf{X}_2^\top + N_2 \lambda_s \mathbf{I}_M)^{-1} (\xi \mathbf{X}_2 \mathbf{y}_2^t + (1 - \xi) \mathbf{X}_2 \mathbf{y}_2), \end{aligned} \quad (3)$$

where λ_s is the student regularization parameter. While it is common to restrict $\xi \in [0, 1]$ (Lopez-Paz et al., 2015), we do not impose this constraint, in line with (Das & Sanghavi, 2023; Pareek et al., 2024). From (3), the parameter ξ is independent of $\mathbf{Q}_2$, making it possible to choose a negative ξ that achieves better generalization performance. For the covariates $\mathbf{x}_j^{(i)}$ and the noise terms $\varepsilon_j^{(i)}$, $i = 1, 2$, $1 \leq j \leq N_i$, we make the following assumptions, which are standard in the random matrix theory literature (see, e.g., (Bai & Silverstein, 2010)).

Assumption 2.1. Suppose $\mathbf{X}_1, \mathbf{X}_2, \varepsilon_1$, and ε_2 are mutually independent. Moreover, we assume

(a) the covariates are generated according to

$$\mathbf{X}_i = (\Sigma_i)^{1/2} \mathbf{Z}_i, \quad \text{for } i = 1, 2,$$

where $\mathbf{Z}_i = (z_{jk}^{(i)})$ is an $M \times N_i$ random matrix with i.i.d. entries of zero mean and unit variance, and Σ_i is a positive semi-definite matrix. Furthermore, we assume for all $p \in \mathbb{N}$, there is a constant C_p such that

$$\max_{i=1,2} \mathbb{E} |z_{11}^{(i)}|^p \leq C_p. \quad (4)$$

(b) $M \sim N_1 \sim N_2$.

(c) $\varepsilon_i \in \mathbb{R}^{N_i}$ is a random vector consisting of i.i.d. entries of zero mean, variance σ^2 , and for all $p \in \mathbb{N}$, there is a constant c_p such that

$$\max_{i=1,2} \mathbb{E} |\varepsilon_1^{(i)}|^p \leq c_p.$$

While we allow $z_{11}^{(1)}$ and $z_{11}^{(2)}$ to follow different distributions – a form of covariate shift – our theoretical results do not depend on their specific distributions, provided that the moment conditions in Assumption 2.1(a) are satisfied. The requirement that all moments of $z_{11}^{(i)}$ exist can be relaxed to the existence of $(8 + c)$ -th moment for any positive constant c , with minor modifications to our proof and hence we do not pursue this generalization here. The following assumption on the structure of the covariance matrices is imposed to facilitate theoretical analysis and rule out degenerate cases.

Assumption 2.2. Let τ be a small constant. Denote the eigenvalues of Σ_i by $\sigma_1^i \geq \sigma_2^i \cdots \geq \sigma_M^i \geq 0$.

(a) (Boundedness of Σ_i). We assume that $\max_{i=1,2} \|\Sigma_i\| = \sigma_1^i < \tau^{-1}$.

(b) (Anti-concentration at 0). For $i = 1, 2$, the empirical spectral distribution of Σ_i satisfies

$$\frac{1}{M} \sum_{j=1}^M \delta(\sigma_j^i \leq \tau) \leq 1 - \tau.$$

Let $(\mathbf{x}, y)$ be an unseen sample of the target task, that is $y = \beta_2^\top \mathbf{x} + \varepsilon$, where $\mathbf{x} \sim D_2$ and ε follows the same distribution with $\varepsilon_1^{(2)}$. Under the mean squared loss, the generalization ability is quantified by the risk of the estimator β_s :

$$\begin{aligned} \mathbf{R}(\beta_s) &= \mathbb{E}_{\mathbf{x}, y} |y - \beta_s^\top \mathbf{x}|^2 = \mathbb{E}_{\mathbf{x}, y} |(\beta_2 - \beta_s)^\top \mathbf{x} + \varepsilon|^2 \\ &= \|\Sigma_2^{1/2}(\beta_2 - \beta_s)\|^2 + \sigma^2, \end{aligned}$$

where $\mathbb{E}_{\mathbf{x}, y}$ denotes the expectation taken with respect to (w.r.t.) the pair $(\mathbf{x}, y)$. The generalization ability of the student model is measured by the excess risk, which is defined as follows:

$$\mathbf{ER}(\beta_s) = \mathbf{R}(\beta_s) - \sigma^2 = \|\Sigma_2^{1/2}(\beta_2 - \beta_s)\|^2. \quad (5)$$

When $\xi = 0$, β_s reduces to the ridge regression estimator for the student only model, and we denote the corresponding excess risk by $\mathbf{ER}_0$. One may easily check that $\mathbf{ER} = O(1)$ almost surely. In the remainder of this paper, we present a bias-variance decomposition of the excess risk using tools from random matrix theory and use the theoretical results to analyze the generalization performance of the student model.

2.2. Random matrix theory

Before proceeding to the theoretical analysis, we introduce several key quantities from random matrix theory that will appear in our main results. For any distribution G supported on $\mathbb{R}^+ = [0, \infty)$, its Stieltjes transform is defined as

$$m_G(z) = \int \frac{1}{x - z} dG(x), \quad z \notin \text{supp}(G).$$

Next, we define the asymptotic eigenvalue density of random matrices via its Stieltjes transform. This lemma is well-known in the random matrix theory literature (e.g., (Bai & Silverstein, 2010)).

Lemma 2.3. Let $\mathbf{X} = \Sigma^{1/2} \mathbf{Z}$ be a random matrix, where $\mathbf{Z} = (z_{jk}) \in \mathbb{R}^{M \times N}$, $M \sim N$ satisfies Assumption 2.1(a), and Σ satisfies Assumption 2.2. For each $z \in \mathbb{C} \setminus \mathbb{R}^+$, there exists a unique $m \equiv m_M(z) \in \mathbb{C}$ satisfying the equation

$$z = -\frac{1}{m} + \frac{1}{N} \text{Tr} \frac{\Sigma}{1 + m\Sigma} = -\frac{1}{m} - \frac{z}{N} \text{Tr} \Sigma \Pi, \quad (6)$$

with $\Im z \Im m(z) \geq 0$, where $\Pi(z) = -(z + z m \Sigma)^{-1}$.

3. Theoretical Analysis

In this section, we analyze the excess risk $\mathbf{ER}(\beta_s)$ defined in 5 under three distinct settings. In Section 3.1, we consider the case where β_1 and β_2 are deterministic, with their difference being arbitrary. In Section 3.2, we study the scenario in which $\beta_1 = \beta_2$ and the common parameter vector is drawn from a prior distribution. Finally, in Section 3.3, we analyze ridgeless regression under the regime where $M < N_1, N_2$ and the covariance matrices Σ_1, Σ_2 are invertible.

Before presenting the main results, we first introduce some necessary notation. For M, N_i, Σ_i and $z < 0$, the Stieltjes transform determined by Lemma 2.3 is denoted by $m_i(z)$.

Let $\Pi_i(z) = -(z + z m_i(z) \Sigma_i)^{-1}$, $i = 1, 2$. We write $\mathbf{Q}_1 = \mathbf{Q}_1(-\lambda_t)$, $\mathbf{Q}_2 = \mathbf{Q}_2(-\lambda_s)$, $\Pi_1 = \Pi_1(-\lambda_t)$ and $\Pi_2 = \Pi_2(-\lambda_s)$ for notational simplicity. For any deterministic matrix $\mathbf{A}$ with bounded spectral norm, we define for $i = 1, 2$,

$$\mathcal{S}_i(\mathbf{A}) = \mathbf{A} + \frac{\frac{1}{N_i} \text{Tr} \Sigma_i \Pi_i \mathbf{A} \Pi_i}{(1 + \frac{1}{N_i} \text{Tr} \Sigma_i \Pi_i)^2 - \frac{1}{N_i} \text{Tr} (\Sigma_i \Pi_i)^2} \Sigma_i.$$

Moreover, when $\mathbf{A} = \mathbf{I}_M$, we denote

$$\Pi'_1 = \frac{d}{dz} \Pi_1(z) \Big|_{z=-\lambda_t} = \Pi_1 \mathcal{S}_1(\mathbf{I}_M) \Pi_1,$$

$$\Pi'_2 = \frac{d}{dz} \Pi_2(z) \Big|_{z=-\lambda_s} = \Pi_2 \mathcal{S}_2(\mathbf{I}_M) \Pi_2.$$

The other quantities are summarized as follow

$$\mathbf{E}_1 = \Pi_1 \mathcal{S}_1(\Sigma_2) \Pi_1, \quad \mathbf{E}_2 = \Pi_1 \mathcal{S}_1(\Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2) \Pi_1,$$

$$\mathbf{E}_3 = \Pi_1 \mathcal{S}_1(\Sigma_2 \Pi_2) \Pi_1, \mathbf{E}_4 = \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2, \mathbf{E}_5 = \Sigma_2 \Pi_2.$$

3.1. Deterministic regression parameters

We now state our first main result.

Theorem 3.1. Let $\gamma = \beta_1 - \beta_2$. For the deterministic vectors $\|\beta_1\|$ and $\|\beta_2\|$, assume that $\|\beta_1\|, \|\beta_2\| \leq c$ for some constant c . Under Assumptions 2.1-2.2, the following results hold:

$$\mathbf{ER}(\beta_s) = \widehat{\mathbf{Bias}} + \widehat{\mathbf{Var}} + o_{a.s.}(1),$$

where

$$\begin{aligned} \widehat{\mathbf{Bias}} &= \xi^2 \beta_1^\top [\lambda_t^2 \mathbf{E}_1 + \lambda_s^2 \lambda_t^2 \mathbf{E}_2 - 2\lambda_t \lambda_s \mathbf{E}_3] \beta_1 \\ &+ \lambda_s^2 \beta_2^\top \mathbf{E}_4 \beta_2 + 2\xi \beta_1^\top [\lambda_t \lambda_s \Pi_1 \mathbf{E}_5 - \lambda_t \lambda_s^2 \Pi_1 \mathbf{E}_4] \beta_2 \\ &+ \xi^2 \gamma^\top [-2\lambda_s \mathbf{E}_5 + \lambda_s^2 \mathbf{E}_4] \gamma + 2\xi \beta_2^\top [\lambda_s^2 \mathbf{E}_4 - \lambda_s \mathbf{E}_5] \gamma \\ &+ 2\xi^2 \lambda_t \lambda_s \gamma^\top [\mathbf{E}_5 \Pi_1 - \lambda_s \mathbf{E}_4 \Pi_1 - \frac{1}{\lambda_s} \Sigma_2 \Pi_1 + \mathbf{E}_5 \Pi_1] \beta_1, \end{aligned} \quad (7)$$

and

$$\begin{aligned} \widehat{\mathbf{Var}} &= \frac{\xi^2 \sigma^2}{N_1} \text{Tr}[(\Sigma_2 - 2\lambda_s \mathbf{E}_5 + \lambda_s^2 \mathbf{E}_4)(\Pi_1 - \lambda_t \Pi_1')] \\ &+ \frac{(1 - \xi)^2 \sigma^2}{N_2} \text{Tr}[\mathbf{E}_5 - \lambda_s \Sigma_2 \Pi_2']. \end{aligned} \quad (8)$$

This theorem characterizes the dependence of $\mathbf{ER}(\beta_s)$ on the geometry of $\Sigma_1, \beta_1, \Sigma_2$, and β_2 . We provide an illustrative example here. Suppose that Σ_i admits the spectral decomposition $\Sigma_i = \mathbf{U}_i \Lambda_i \mathbf{U}_i^\top$, for $i = 1, 2$. Consider the term $\beta_1^\top \Pi_1 \mathbf{E}_5 \beta_2$, which can be expressed as

$$\begin{aligned} &\beta_1^\top (\lambda_t + \lambda_t m_1 \Sigma_1)^{-1} (\lambda_s + \lambda_s m_2 \Sigma_2)^{-1} \Sigma_2 \beta_2 \\ &= (\lambda_s \lambda_t)^{-1} \tilde{\beta}_1^\top (1 + m_1 \Lambda_1)^{-1} \mathbf{U}_1^\top \mathbf{U}_2 \tilde{\Lambda}_2 \tilde{\beta}_2, \end{aligned} \quad (9)$$

where $\tilde{\Lambda}_2$ is a diagonal matrix with entries $\tilde{\Lambda}_{2,jj} = \frac{\Lambda_{2,jj}}{1 + m_2 \Lambda_{2,jj}}$. The vector $\tilde{\beta}_i = \mathbf{U}_i \beta_i$ captures the alignment between β_i and the eigenvectors of Σ_i . The right-hand side of (9) explicitly reveals how the term $\beta_1^\top \Pi_1 \mathbf{E}_5 \beta_2$ depends on $\tilde{\beta}_i$, the eigenvalues of Σ_1 and Σ_2 , and the eigenvector overlap $\mathbf{U}_1^\top \mathbf{U}_2$ between the two covariance matrices. In the special case where each β_i is aligned with an eigenvector of Σ_i – for simplicity, suppose it corresponds to the first eigenvector – the expression (9) further simplifies to $\beta_1^\top \beta_2 (\lambda_s \lambda_t)^{-1} \frac{\Lambda_{2,11}}{(1 + m_1 \Lambda_{1,11})(1 + m_2 \Lambda_{2,11})}$, which depends on the eigenvalues of Σ_i and the inner product $\beta_1^\top \beta_2$. This observation extends the results of (Hastie et al., 2022), which considers high-dimensional least squares regression within a single domain (corresponding to $\xi = 0$ in (3)).

3.2. Random Regression parameters

In this section, we assume that the vector $\beta_1 = \beta_2 = \beta$ is random, and consider the excess risk under two population covariance matrices, Σ_1 and Σ_2 , which may be equal or distinct. Before presenting the main result, we introduce the following assumption, commonly used in the literature (Dobriban & Wager, 2018; Moniri & Hassani, 2025).

Assumption 3.2. The regression parameter vector $\beta = (\beta_1, \dots, \beta_M)^\top \in \mathbb{R}^M$ is random, with each entry i.i.d., and β_1 satisfies

$$\mathbb{E}\beta_1 = 0, \quad \mathbb{E}\beta_1^2 = \frac{\tilde{\sigma}^2}{M}, \quad \text{and } \mathbb{E}|\sqrt{M}\beta_1|^p \leq C_p,$$

for any $p \in \mathbb{N}$, where C_p is a constant depending only on p .

Theorem 3.3. Suppose Assumptions 2.1-3.2 hold. Then the following asymptotic expressions hold:

$$\mathbf{ER}(\beta_s) = \widehat{\mathbf{Bias}} + \widehat{\mathbf{Var}} + o_{a.s.}(1),$$

where

$$\begin{aligned} \widehat{\mathbf{Bias}} &= \frac{\tilde{\sigma}^2}{M} \left[\xi^2 \lambda_t^2 \text{Tr} \Sigma_2 \Pi_1' + 2\xi \lambda_t \lambda_s \text{Tr} \Pi_1 \Pi_2 \Sigma_2 \right. \\ &+ \lambda_s^2 \text{Tr} \Sigma_2 \Pi_2' - 2\xi^2 \lambda_t^2 \lambda_s \text{Tr} \Sigma_2 \Pi_2 \Pi_1' \\ &\left. + \xi \lambda_t \lambda_s^2 \text{Tr} [\mathbf{E}_4 (-2\Pi_1 + \xi \lambda_t \Pi_1')] \right], \end{aligned}$$

and $\widehat{\mathbf{Var}}$ coincides with the expression in Theorem 3.1.

This theorem extends the result of (Moniri & Hassani, 2025), which considers the case of no covariate shift, with inputs drawn i.i.d. from $\mathcal{N}(0, \mathbf{I}_M)$ in the context of W2S generalization (i.e., when $\xi = 1$). Our framework generalizes this analysis by allowing $\xi \in \mathbb{R}$, thereby providing a more comprehensive understanding of the trade-off between learning from the teacher and from the observed labels.

Let $\underline{m}_1(z), \underline{m}_2(z)$ be the Stieltjes transforms of the standard Marchenko-Pastur law with parameters $M/N_1, M/N_2$, respectively; their explicit forms are given in (33). The following corollary follows immediately from Theorem 3.3 and the fact that $\Pi_1 = \underline{m}_1 \mathbf{I}_M, \Pi_2 = \underline{m}_2 \mathbf{I}_M$ (see, e.g., (Alex et al., 2014)).

Corollary 3.4. Suppose $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$. Write $\underline{m}_1 = \underline{m}_1(-\lambda_t), \underline{m}_2 = \underline{m}_2(-\lambda_s)$. Under Assumption 2.1 and Assumption 3.2, we have the following expressions:

$$\begin{aligned} \widehat{\mathbf{Bias}} &= \tilde{\sigma}^2 [\xi^2 \lambda_t^2 \underline{m}_1' + 2\xi \lambda_t \lambda_s \underline{m}_1 \underline{m}_2 + \lambda_s^2 \underline{m}_2' \\ &- 2\xi^2 \lambda_t^2 \lambda_s \underline{m}_2 \underline{m}_1' - 2\xi \lambda_t \lambda_s^2 \underline{m}_2' \underline{m}_1 + \xi^2 \lambda_t^2 \lambda_s^2 \underline{m}_1' \underline{m}_2'], \end{aligned}$$

and

$$\begin{aligned} \widehat{\mathbf{Var}} &= \xi^2 \sigma^2 \frac{M}{N_1} [\underline{m}_1 - 2\lambda_s \underline{m}_1 \underline{m}_2 + \lambda_s^2 \underline{m}_1 \underline{m}_2' - \lambda_t \underline{m}_1' \\ &+ 2\lambda_t \lambda_s \underline{m}_2 \underline{m}_1' - \lambda_t \lambda_s^2 \underline{m}_1' \underline{m}_2'] \\ &+ (1 - \xi)^2 \sigma^2 \frac{M}{N_2} [\underline{m}_2 - \lambda_s \underline{m}_2'], \end{aligned}$$

where $\underline{m}_1', \underline{m}_2'$ denote the derivatives evaluated at $z = -\lambda_t$ and $z = -\lambda_s$, respectively:

$$\underline{m}_1' = \frac{d}{dz} \underline{m}_1(z) \Big|_{z=-\lambda_t}, \quad \underline{m}_2' = \frac{d}{dz} \underline{m}_2(z) \Big|_{z=-\lambda_s}.$$

As previously noted, we do not restrict ξ to the interval $[0, 1]$. It has been shown in (Das & Sanghavi, 2023) that the optimal value of ξ may exceed 1. In Corollary 3.5 below, we present a toy example demonstrating that even when the input data across domains are i.i.d. and in the absence of model shift – i.e., with no domain shift – the limiting optimal value of ξ can be negative.

Corollary 3.5. Suppose the conditions in Corollary 3.4 hold. The limiting optimal value of $\xi < 0$ if

$$\lambda_s \lambda_t \underline{m}_1 \text{SNR} - \frac{M}{N_2} > 0, \quad (10)$$

where $\text{SNR} = \frac{\hat{\sigma}^2}{\sigma^2} = \frac{\|\beta\|^2}{\sigma^2} + o_{a.s.}(1)$.

Remark 3.6. We call the case $\xi < 0$ **anti-learning against the teacher's supervision**, in contrast to $\xi > 1$, which (Das & Sanghavi, 2023) termed anti-learning the observed (possibly noisy) labels. This corollary provides insight into the selection of ξ : the sign of the limiting optimal value of ξ depends not only on parameters (λ_t, λ_s) but also on data-related factors (SNR, data dimension, and sample sizes of both domains).

3.3. Ridgeless Regression

In this section, we consider the minimum ℓ_2 norm least squares (ridgeless) regression estimator. Specifically, the teacher model is defined by

$$\beta_t = (\mathbf{X}_1 \mathbf{X}_1^\top)^+ \mathbf{X}_1 \mathbf{y}_1,$$

where $(\mathbf{X}_1 \mathbf{X}_1^\top)^+$ denotes the Moore-Penrose inverse of $\mathbf{X}_1 \mathbf{X}_1^\top$. Similarly, the ridgeless estimator of β_s takes the form

$$\beta_s = (\mathbf{X}_2 \mathbf{X}_2^\top)^+ [\xi \mathbf{X}_2 \mathbf{X}_2^\top \beta_t + (1 - \xi) \mathbf{X}_2 \mathbf{y}_2].$$

Theorem 3.7. (1) Suppose β_1, β_2 are deterministic, and Assumptions 2.1-2.2 hold. We further assume that for $i = 1, 2$,

$$\left| \frac{M}{N_i} - 1 \right| \geq \tau, \quad \tau \leq \sigma_{\min}(\Sigma_i) \leq \dots \leq \sigma_{\max}(\Sigma_i) \leq \tau^{-1}.$$

Define $f(\lambda) = \widehat{\text{Bias}}$ and $g(\lambda) = \widehat{\text{Var}}$, with $\lambda = \lambda_s = \lambda_t$, where the expressions for $\widehat{\text{Bias}}$ and $\widehat{\text{Var}}$ are provided in (7) and (8), respectively. We have

$$\mathbf{ER}(\beta_s) = f(0^+) + g(0^+) + o_{a.s.}(1). \quad (11)$$

(2) Suppose $\beta = \beta_1 = \beta_2$ are random and Assumptions 2.1-3.2 hold. Then, the estimated expressions in (11) still holds with $f(\lambda) = \widehat{\text{Bias}}$ replaced by the $\widehat{\text{Bias}}$ defined in Theorem 3.3.

If a matrix $\mathbf{A}$ is nonsingular, $\mathbf{A}^+ = \mathbf{A}^{-1}$. The following corollary gives the characterization of $\mathbf{ER}(\beta_s)$ in the under-parameterized setting.

Corollary 3.8. Suppose the conditions in Theorem 3.7 hold and $\frac{M}{N_1}, \frac{M}{N_2} \leq 1 - \tau$. The estimator for student model obtained by (1) is the averaging estimator:

$$\beta_s = \xi \beta_1^{\text{OLS}} + (1 - \xi) \beta_2^{\text{OLS}}, \quad (12)$$

where $\beta_i^{\text{OLS}} = (\mathbf{X}_i \mathbf{X}_i^\top)^{-1} \mathbf{X}_i \mathbf{y}_i$, $i = 1, 2$. Adopting the notation $\gamma = \beta_1 - \beta_2$ in Theorem 3.1, we have

$$\mathbf{ER}(\beta_s) = \widehat{\text{Bias}} + \widehat{\text{Var}} + o_{a.s.}(1),$$

where

$$\widehat{\text{Bias}} = \xi^2 \gamma^\top \Sigma_2 \gamma,$$

and

$$\widehat{\text{Var}} = (1 - \xi)^2 \sigma^2 \frac{M}{N_2 - M} + \xi^2 \sigma^2 \frac{1}{N_1 - M} \text{Tr} \Sigma_2 \Sigma_1^{-1}.$$

Based on the conclusions of Theorems 3.1-3.7, the high-dimensional asymptotic excess risk, regarded as a function of ξ , is a quadratic function. Given that the excess risk is non-negative, the quadratic function opens upwards. This observation is consistent with (Pareek et al., 2024), where self-distillation is considered. Given a $\xi \in \mathbb{R}$, the gain of cross-domain KD is characterized by the reduction in excess risk, $\mathbf{ER}_0 - \mathbf{ER}(\beta_s)$.

Proposition 3.9. Under the conditions of Theorem 3.1 and Assumption B.10 for the deterministic case in Appendix B.8, there exists a value of $\xi \in \mathbb{R}$ such that

$$\min_{\xi \in \mathbb{R}} (\mathbf{ER}(\beta_s) - \mathbf{ER}_0) < 0. \quad a.s. \quad (13)$$

Moreover, under the conditions of Theorem 3.1 and Assumption B.11 for the random case in Appendix B.8, the inequality 13 also holds.

Remark 3.10. This proposition shows that, even in the presence of a significant domain discrepancy, it is possible to find a value of $\xi \in \mathbb{R}$ such that the student model outperforms the student-only baseline (i.e., training on the observed labels only). We provide further details in Appendix B.8, where we provide closed-form expressions for the optimal ξ^* under several common settings and demonstrate that covariate shift can, in some cases, be beneficial for KD.

3.4. Numerical Simulations

We plot the excess risk of the student model: (a) under model shift with identical covariate distributions, and (b) under covariate shift with identical parameter vectors, in Figure 1. All theoretical values of the Stieltjes transform presented in this paper are obtained by solving (6). Due to space limitations, the numerical validation of Corollary 3.8 is provided in Appendix C.3. Simulation results, averaged over 100 independent trials, show good agreement with the theoretical predictions.

Furthermore, we present numerical simulations of $\mathbf{ER}(\beta_s)$ as a function of λ_s and λ_t for various values of ξ ; these results are included in Appendix C.1.

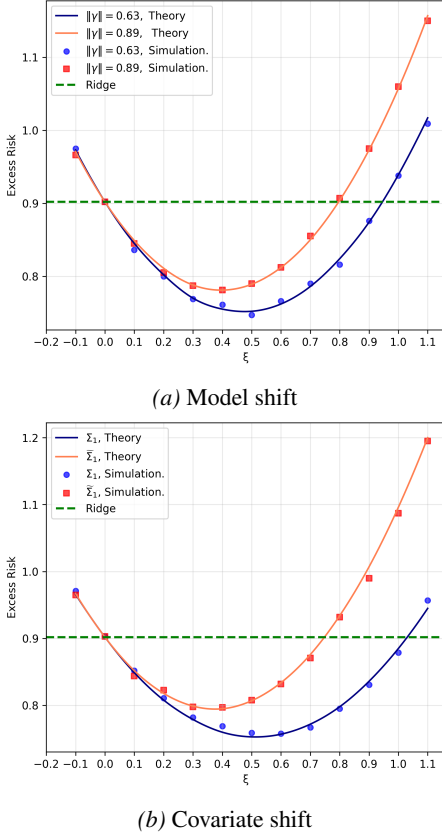


Figure 1. Excess risk in the presence of domain shift. Solid lines represent theoretical values, while scattered points denote simulation results (averaged over 100 trials). The dashed green line indicates the theoretical performance for student-only baseline, corresponding to ridge regression trained solely on the target domain data. (a) Settings: $(\lambda_t, \lambda_s) = (0.1, 0.5)$, $(M, N_1, N_2) = (400, 600, 200)$, $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$. The vectors $\beta_2 = (1, \dots, 1)^T / \sqrt{M}$, $\sigma^2 = 1$. We label the case $\|\gamma\| = 0.63$ as $\gamma = -(2, \dots, 2, 0, \dots, 0)^T / \sqrt{M}$ with the first $M/10$ entries equal to $-2/\sqrt{M}$, and the case $\|\gamma\| = 0.89$ with the first $M/5$ entries equal to $-2/\sqrt{M}$. (b) Settings: $(\lambda_t, \lambda_s) = (0.1, 0.5)$, $\beta_1 = \beta_2 \sim \mathcal{N}(0, M^{-1}\mathbf{I}_M)$, $(M, N_1, N_2) = (600, 200, 300)$, $\Sigma_1 = 4\mathbf{I}_M$, $\tilde{\Sigma}_1 = \text{diag}(d_1, \dots, d_M)$ with $d_i = 0.64\delta(i \leq M/2) + 0.25\delta(M/2 < i \leq M)$, $\sigma^2 = 1$.

4. Non-monotone excess risk

In this section, fixing ξ , λ_t and λ_s , we examine the excess risk as a function of the dimension M and the sample sizes N_1 and N_2 . We find that the student model exhibits the double descent phenomenon, characterized by a non-monotonic behavior of the excess risk as a function of the ratio of dimension-to-sample-size. This phenomenon is consistent with findings in various linear regression settings (Hastie et al., 2022; Nakkiran et al., 2021; Belkin et al., 2020; Moniri & Hassani, 2025), and has been previously observed by (Moniri & Hassani, 2025) in the special case of pure teacher supervision without domain shift, where the risk was studied as a function of $\frac{M}{N_1}$.

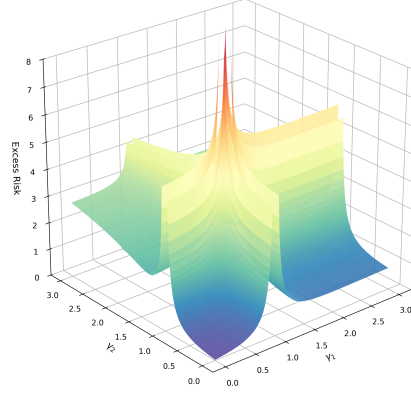


Figure 2. Crossed double decent phenomenon: asymptotic excess risk as a function of γ_1 and γ_2 for the ridgeless regression. Settings: $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$, $\sigma^2 = 1$, $\xi = 0.5$, $\text{SNR}=2$, $\tau = 0.04$.

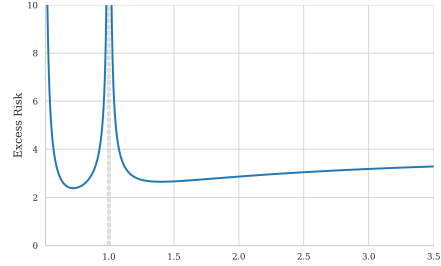


Figure 3. Asymptotic excess risk curve for ridgeless regression as a function of γ_1 . Settings: $\xi = 0.5$, $\gamma_1 - \gamma_2 = -0.5$, $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$, $\sigma^2 = 1$, $\text{SNR}=2$.

We consider the specific case of ridgeless regression where $\beta_1 = \beta_2 = \beta$ is a random vector satisfying Assumptions 2.1–3.2, with isotropic features $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$. We denote $\gamma_1 = \frac{M}{N_1}$, $\gamma_2 = \frac{M}{N_2}$. The theoretical results, which characterize the asymptotic excess risk in terms of its bivariate dependence on γ_1 and γ_2 , are provided in the Appendix B.6. Specifically, we identify that the interaction between these two ratios gives rise to a *crossed double descent* phenomenon, as shown in Figure 2. Notably, the asymptotic excess risk exhibits a pattern of more than one descent in certain regimes. To visualize this behavior, we plot the risk trajectories for the specific cases of $\gamma_1 - \gamma_2 = 0.5$ and $\gamma_1 - \gamma_2 = -0.5$ in Figures 3 and 4.

5. Extension

5.1. Nonlinear case

Our theoretical results are initially established for linear models; however, we anticipate that they can be extended to more general settings. To explore this extension, we conduct numerical simulations specifically for nonlinear models here. We assume the source data $\{(\mathbf{x}_j^{(1)}, y_j^{(1)})\}_{j=1}^{N_1}$ are generated

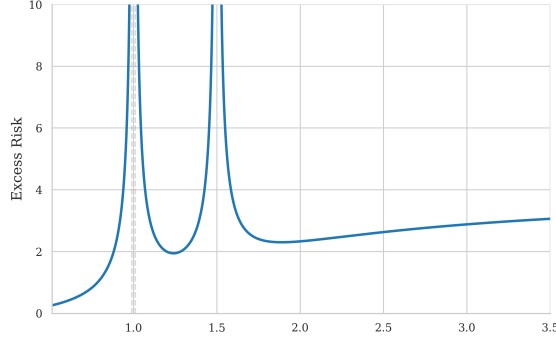


Figure 4. Asymptotic excess risk curve for ridgeless regression as a function of γ_1 . Settings: $\xi = 0.5$, $\gamma_1 - \gamma_2 = 0.5$, $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$, $\sigma^2 = 1$, $\text{SNR}=2$.

i.i.d. according to $y_j^{(1)} = f(\mathbf{x}_j^{(1)}) + \varepsilon_j^{(1)}$, for $1 \leq j \leq N_1$. The target data $\{(\mathbf{x}_j^{(2)}, y_j^{(2)})\}_{j=1}^{N_2}$ are generated according to $y_j^{(2)} = \tilde{f}(\mathbf{x}_j^{(2)}) + \varepsilon_j^{(2)}$, for $1 \leq j \leq N_2$. Suppose $\mathbf{x}_j^{(1)} \sim D_1$, $1 \leq j \leq N_1$ and $\mathbf{x}_j^{(2)} \sim D_2$, $1 \leq j \leq N_2$. We refer to the case $D_1 \neq D_2$ as a covariate shift, and the case where $f \neq \tilde{f}$ as a model shift.

We consider learning the unknown function using a fully connected two-layer neural network with n hidden neurons: $f_{\text{NN}}(\mathbf{x}) = \mathbf{a}^\top \sigma(\mathbf{W}\mathbf{x})$, where $\mathbf{W} \in \mathbb{R}^{n \times M}$ is the weight matrix, and $\sigma(\cdot)$ is an activation function applied entrywise. When the random weight matrix $\mathbf{W}$ is fixed and only the second-layer weight $\mathbf{a}$ is optimized, the model reduces to a kernel regression model, where the kernel defined by $\mathbf{x} \rightarrow \sigma(\mathbf{W}\mathbf{x})$ is referred to as the conjugate kernel (Neal, 2012). The teacher model is given by $f_{\text{NN}}^t(\mathbf{x}) = \mathbf{a}_t^\top \sigma(\tilde{\mathbf{W}}_1 \mathbf{x})$, with $\mathbf{a}_t = \arg \min_{\mathbf{a}} \frac{1}{N_1} \|\mathbf{y}_1 - \sigma(\mathbf{X}_1^\top \tilde{\mathbf{W}}_1^\top) \mathbf{a}\|^2 + \lambda_t \|\mathbf{a}\|^2$. We use $f_{\text{NN}}^t(\mathbf{x}_j^{(2)})$ together with the covariates $\{\mathbf{x}_j^{(2)}\}_{j=1}^{N_2}$ to generate predictions $\mathbf{y}_2^t$. Then the student model is finetuned on the target domain data and $\mathbf{y}_2^t$. The student model takes the form $f_{\text{NN}}^s(\mathbf{x}) = \mathbf{a}_s^\top \sigma(\mathbf{W}_1 \mathbf{x})$ with

$$\begin{aligned} \mathbf{a}_s = \arg \min_{\mathbf{a}} & \xi \left(\frac{1}{N_2} \|\mathbf{y}_2^t - \sigma(\mathbf{X}_2^\top \mathbf{W}_1^\top) \mathbf{a}\|^2 \right) \\ & + (1 - \xi) \left(\frac{1}{N_2} \|\mathbf{y}_2 - \sigma(\mathbf{X}_2^\top \mathbf{W}_1^\top) \mathbf{a}\|^2 \right) + \lambda_s \|\mathbf{a}\|^2. \end{aligned}$$

We also examine a setting where the teacher model is a deeper neural network. Specifically, while keeping the student model fixed, we let the teacher be a 4-layer fully connected network:

$$f_{\text{NN}}^t = \mathbf{a}_t^\top \sigma(\tilde{\mathbf{W}}_3 \sigma(\tilde{\mathbf{W}}_2 \sigma(\tilde{\mathbf{W}}_1 \mathbf{x}))),$$

where $\mathbf{a}_t$ is the solution to

$$\arg \min_{\mathbf{a}} \frac{1}{N_1} \|\mathbf{y}_1 - [\sigma(\tilde{\mathbf{W}}_3 \sigma(\tilde{\mathbf{W}}_2 \sigma(\tilde{\mathbf{W}}_1 \mathbf{X}_1))]^\top \mathbf{a}\|^2 + \lambda_t \|\mathbf{a}\|^2.$$

We set $D_1 = \mathcal{N}(0, 4\mathbf{I}_M)$ and $D_2 = \mathcal{N}(0, \mathbf{I}_M)$. Let $f(\mathbf{x}) = (\beta^\top \mathbf{x})^2 + 1$, $\tilde{f}(\mathbf{x}) = (\beta^\top \mathbf{x})^2$. Because $D_1 \neq D_2$ and $f \neq \tilde{f}$, both covariate shift and model shift are present in this setting. More details and the numerical results are provided in Appendix C.2.

5.2. Dependence between domains

In this section, we consider two cases in which $\mathbf{X}_1$ and $\mathbf{X}_2$ are not fully independent.

Case 1: Assume $\mathbf{X}_1$ exhibits weak dependence on $\mathbf{X}_2$ in the following sense: $\mathbf{X}_1 = \alpha \mathbf{X}_2 + \tilde{\mathbf{X}}_1$, where $\alpha \rightarrow 0$ as $M \rightarrow \infty$, and $\tilde{\mathbf{X}}_1$ is independent of $\mathbf{X}_2$ and takes the form $\Sigma_1^{1/2} \mathbf{Z}_1$. It is easy to see $\text{Cov}(\mathbf{x}_j^{(1)}, \mathbf{x}_j^{(2)}) = \text{Cov}(\mathbf{x}_j^{(2)}, \alpha \mathbf{x}_j^{(2)}) = \alpha \Sigma_2$.

Case 2: Suppose $\mathbf{X}_1$ is a signal-plus-noise data matrix: $\mathbf{X}_1 = \mathbf{X}_2 + \mathbf{A}$, where $\mathbf{A}$ is a deterministic signal matrix with $\|\mathbf{A}\| = o(\sqrt{M})$. This model captures realistic scenarios in domain adaptation where the source and target domains share a common underlying data matrix, but differ by a small deterministic shift—such as a faint shared signal across features in source domain.

In *Case 1*, our theoretical analysis remains valid. For *Case 2*, we obtain a new limiting behavior; the theoretical results and technical details are provided in Section B.10.

5.3. Multi-source data

Extending our framework to multi-source settings with multiple teacher models is a natural generalization of our work. For illustrative simplicity, we present the two-teacher case, where the teachers are derived from distinct domains. Let the two teacher models have parameters β_{10} and β_{11} , trained on datasets $(\mathbf{X}_0, \mathbf{y}_0)$ and $(\mathbf{X}_1, \mathbf{y}_1)$, respectively. Their predictions on the target domain data $\mathbf{X}_2$ are denoted as $\mathbf{y}_{10} = \mathbf{X}_2^\top \beta_{10}$ and $\mathbf{y}_{11} = \mathbf{X}_2^\top \beta_{11}$. The extended loss function is then given by

$$\begin{aligned} \ell(\beta) = & \frac{\xi_1}{N_2} \|\mathbf{y}_{10} - \mathbf{X}_2^\top \beta\|^2 + \frac{\xi_2}{N_2} \|\mathbf{y}_{11} - \mathbf{X}_2^\top \beta\|^2 \\ & + \frac{1 - \xi_1 - \xi_2}{N_2} \|\mathbf{y}_2 - \mathbf{X}_2^\top \beta\|^2 + \lambda_s \|\beta\|^2, \end{aligned}$$

which yields the student model

$$\begin{aligned} \beta_s = & (\mathbf{X}_2 \mathbf{X}_2^\top + N_2 \lambda_s \mathbf{I}_M)^{-1} [\xi_1 \mathbf{X}_2 \mathbf{X}_2^\top \beta_{10} + \xi_2 \mathbf{X}_2 \mathbf{X}_2^\top \beta_{11} \\ & + (1 - \xi_1 - \xi_2) \mathbf{X}_2 (\mathbf{X}_2^\top \beta + \varepsilon_2)]. \end{aligned}$$

Denote by β the true parameter vector of the target domain. We aim to derive the limit of $(\beta_s - \beta)^\top \Sigma_2 (\beta_s - \beta)$. The multi-teacher formulation introduces cross-terms of the form

$$\xi_1 \xi_2 \mathbf{F}_0^\top \Sigma_2 \mathbf{F}_1,$$

where

$$\mathbf{F}_i = \left(\frac{1}{N_2} \mathbf{Q}_2 \mathbf{X}_2 \mathbf{X}_2^\top \beta_{1i} - \beta \right), \quad i = 0, 1.$$

Substituting $\beta_{ti} = \frac{1}{N_i} \mathbf{Q}_i \mathbf{X}_i (\mathbf{X}_i^\top \beta_i + \varepsilon_i)$ for $i = 0, 1$ and omitting constants, this term comprises products of

$$\beta, \beta_0, \beta_1, \Sigma_2, \mathbf{Q}_2, \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_2, \\ \mathbf{Q}_0, \mathbf{Q}_1, \mathbf{Q}_0 \mathbf{X}_0 \varepsilon_0, \mathbf{Q}_1 \mathbf{X}_1 \varepsilon_1,$$

where β_0 and β_1 are the parameter vectors of the respective teacher models. Under the assumption of domain independence, the limiting behavior of these terms can be established by applying the same arguments presented in the Appendix. For scenarios involving more than two teachers, the derivation remains analogous; one only needs to include additional cross-terms with similar functional forms.

6. Conclusion

This paper provides a precise theoretical characterization of the asymptotic excess risk in cross-domain KD using random matrix theory and bias-variance decomposition. We demonstrate that for any real-valued imitation parameter ξ , distillation can outperform student only training even under significant domain discrepancies, highlighting its efficacy in transferring knowledge across domains. While our current analysis focuses on linear models and observes a novel double descent phenomenon, future work should extend these theoretical frameworks to nonlinear architectures to further generalize these findings and provide a rigorous characterization of their risk landscapes.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. You *can* have an appendix here.

B. Proofs

B.1. Basic Tools

Preliminary definitions and auxiliary lemmas used in the proofs of the main results are provided in this section.

Lemma B.1. (Lemma B.26 in (Bai & Silverstein, 2010)) Let $\mathbf{C}$ be an $M \times M$ deterministic matrix and $\mathbf{z} = (z_1, \dots, z_M)^\top \in \mathbb{R}^M$ be a random vector of independent entries. Assume that $\mathbb{E}z_i = 0$, $\mathbb{E}|z_i|^2 = 1$, and $\mathbb{E}|z_i|^\ell \leq C_\ell$. Then for any $\ell \geq 1$,

$$\mathbb{E}|\mathbf{z}^\top \mathbf{C} \mathbf{z} - \text{Tr} \mathbf{C}|^\ell \leq c_\ell ((C_4 \text{Tr} \mathbf{C} \mathbf{C}^*)^{\ell/2} + C_{2\ell} \text{Tr}(\mathbf{C} \mathbf{C}^*)^{\ell/2}),$$

where c_ℓ is a constant depending on ℓ only.

Before stating the subsequent results, it is convenient to introduce the notion of stochastic domination.

Definition B.2. Let $\chi = \chi^{(p)}$, $\zeta = \zeta^{(p)}$ be two families of p -dependent random variables. We say that χ is stochastically dominated by ζ if for all small $c > 0$ and large constant $\ell > 0$,

$$\mathbb{P}(|\chi^{(p)}| > p^c |\zeta^{(p)}|) \leq p^{-\ell}$$

for all large p . If χ is stochastically dominated by ζ , we use the notation $\chi \prec \zeta$ or $\chi = O_{\prec}(\zeta)$. We say an event $\mathcal{E}_p$ holds with high probability if

$$\mathbb{P}(\mathcal{E}_p^C) \leq p^{-\ell} \text{ for any fixed } \ell > 0.$$

Lemma B.3. (Lemma 22 in (Yang et al., 2025)) Let $\mathbf{Z}$ satisfies Assumption 2.1(a)-(b). Suppose $\frac{M}{N} \leq 1 - \tau$. Then we have

$$(\sqrt{M} - \sqrt{N})^2 + O_{\prec}(\sqrt{N}) \leq \lambda_{\min}(\mathbf{Z} \mathbf{Z}^\top) \leq \lambda_{\max}(\mathbf{Z} \mathbf{Z}^\top) \leq (\sqrt{M} + \sqrt{N})^2 + O_{\prec}(\sqrt{N}).$$

The upper bound on $\lambda_{\max}(\mathbf{Z} \mathbf{Z}^\top)$ still holds without the assumption $M/N \leq 1 - \tau$.

Lemma B.4. (Corollary 25 in (Yang et al., 2025)) Suppose $\varepsilon_1, \dots, \varepsilon_t$ are independent random vectors satisfying Assumption 2.1(c). Then, we have that for any deterministic vector $\mathbf{v} \in \mathbb{R}^N$,

$$|\mathbf{v}^\top \varepsilon_i| \prec \sigma \|\mathbf{v}\|, \quad i = 1, \dots, t,$$

and for any deterministic matrix $\mathbf{B} \in \mathbb{R}^{N \times N}$,

$$|\varepsilon_i^\top \mathbf{B} \varepsilon_j - \delta(i=j) \sigma^2 \text{Tr} \mathbf{B}| \prec \sigma^2 \|\mathbf{B}\|_F, \quad \text{for } i, j \in [t].$$

Moreover, for any deterministic vector $\mathbf{v}$, we have

$$|\mathbf{v}^\top \varepsilon_i| \prec \sigma \|\mathbf{v}\|, \quad i \in [t].$$

Definition B.5. Let $\mathbf{A}_p, \mathbf{B}_p \in \mathbb{R}^{p \times p}$ be sequences of random or deterministic symmetric real matrices. We say $\mathbf{A}_p, \mathbf{B}_p$ are equivalent, denoted by $\mathbf{A}_p \asymp \mathbf{B}_p$, if

$$\frac{1}{p} \text{Tr} \mathbf{D}_p(\mathbf{A}_p - \mathbf{B}_p) = o_{a.s.}(1) \text{ and } \mathbf{u}^\top (\mathbf{A}_p - \mathbf{B}_p) \mathbf{v} = o_{a.s.}(1)$$

for any sequence of deterministic matrices $\mathbf{D}_p$ and all deterministic vectors $\mathbf{u}, \mathbf{v}$ such that

$$\limsup_p \|\mathbf{D}_p\| < \infty, \quad \limsup_p \max\{\|\mathbf{u}\|, \|\mathbf{v}\|\} < \infty.$$

Lemma B.6. (1) (Theorem 2.6 in (Couillet & Liao, 2022)) We denote by ϱ the probability measure associated with m determined in Lemma 2.3. Let $\mathbf{X} = \Sigma^{1/2} \mathbf{Z} \in \mathbb{R}^{M \times N}$, where the entries of $\mathbf{Z}$ are i.i.d. centered random variables with unit variance and finite $8 + c$ -th moment, for any positive constant c . Suppose Assumption 2.1(b) and Assumption 2.2 hold. Then for $z \in \mathbb{C} \setminus \mathbb{R}^+$, we have

$$\mathbf{Q}(z) \asymp \mathbf{\Pi}(z), \quad \underline{\mathbf{Q}}(z) \asymp m(z) \mathbf{I}_M, \quad (14)$$

where

$$\mathbf{Q}(z) = \left(\frac{\mathbf{X} \mathbf{X}^\top}{N} - z \mathbf{I}_M \right)^{-1}, \quad \underline{\mathbf{Q}}(z) = \left(\frac{\mathbf{X}^\top \mathbf{X}}{N} - z \mathbf{I}_M \right)^{-1}, \quad \mathbf{\Pi}(z) = -(z \mathbf{I}_M + z m \Sigma)^{-1}.$$

(2) When $\frac{M}{N} < 1 - \tau$, (14) still holds at $z = 0$.

Proof: Let $\mathbf{u}, \mathbf{v}$ be two deterministic unit vectors and $f_N(\lambda) = \mathbf{u}^\top \mathbf{Q}(-\lambda) \mathbf{v}$ for $\lambda < 0$. Since $\lambda_{\min}(\frac{\mathbf{X}\mathbf{X}^\top}{N}) > \frac{1}{2}(1 - \sqrt{M/N})^2$ with high probability, we have $|f_N(\lambda)| \leq \|\mathbf{Q}(-\lambda)\| \lesssim 1$, $|f'_N(\lambda)| \leq \|\frac{d}{d\lambda} \mathbf{Q}(-\lambda)\| \lesssim 1$ with high probability. Therefore, $\{f_N(\lambda)\}$ is equicontinuous with high probability. By applying the Arzela-Ascoli theorem, f_N converges uniformly to its limit $f(\lambda) = \mathbf{u}^\top \mathbf{\Pi}(-\lambda) \mathbf{v}$. By the Moore-Osgood theorem, we can exchange limits and get

$$\lim_{N \rightarrow \infty} f_N(0) = \lim_{N \rightarrow \infty} \lim_{\lambda \rightarrow 0^-} f_N(\lambda) = \lim_{\lambda \rightarrow 0^-} \lim_{N \rightarrow \infty} f_N(\lambda) = \lim_{\lambda \rightarrow 0^-} \mathbf{u}^\top \mathbf{\Pi}(-\lambda) \mathbf{v} = \mathbf{u}^\top \mathbf{\Pi}(0) \mathbf{v}, \text{ a.s.},$$

where we use the fact that both $\mathbf{Q}$ and $\mathbf{\Pi}$ are analytic in an open neighborhood of 0 with high probability. Similarly, we can derive $\frac{1}{M} \text{Tr} \mathbf{A} \mathbf{Q}(0) = \frac{1}{M} \text{Tr} \mathbf{A} \mathbf{\Pi}(0) + o_{a.s.}(1)$, which completes the proof.

Lemma B.6 shows that $\mathbf{\Pi}(z)$ is a deterministic equivalent of $\mathbf{Q}(z)$. For technical reasons, we further require the following result.

Lemma B.7. Suppose the conditions in Lemma B.6 hold. $\mathbf{A}$ denotes a deterministic $M \times M$ matrix with bounded spectral norm. For any fixed complex numbers $\tilde{z}_1, \tilde{z}_2 \in \mathbb{C} \setminus \mathbb{R}^+$, we have for all deterministic vectors $\mathbf{u}, \mathbf{v}$,

$$\mathbf{u}^\top (\mathbf{Q}(\tilde{z}_1) \mathbf{A} \mathbf{Q}(\tilde{z}_2) - \mathbf{\Pi}(\tilde{z}_1) \mathbf{S}(\mathbf{A}) \mathbf{\Pi}(\tilde{z}_2)) \mathbf{v} = o_{a.s.}(\|\mathbf{u}\| \|\mathbf{v}\|), \quad (15)$$

where

$$\mathbf{S}(\mathbf{A}) = \mathbf{A} + \frac{\frac{1}{N} \text{Tr} \mathbf{\Sigma} \mathbf{\Pi}(\tilde{z}_1) \mathbf{A} \mathbf{\Pi}(\tilde{z}_2)}{(1 + \frac{1}{N} \text{Tr} \mathbf{\Sigma} \mathbf{\Pi}(\tilde{z}_2))(1 + \frac{1}{N} \text{Tr} \mathbf{\Sigma} \mathbf{\Pi}(\tilde{z}_1)) - \frac{1}{N} \text{Tr} \mathbf{\Sigma} \mathbf{\Pi}(\tilde{z}_1) \mathbf{\Sigma} \mathbf{\Pi}(\tilde{z}_2)} \mathbf{\Sigma}. \quad (16)$$

Moreover, for any deterministic matrix $\mathbf{C} \in \mathbb{R}^{M \times M}$ satisfying $\|\mathbf{C}\| \leq C$ for some constant C , we have

$$\frac{1}{M} \text{Tr} \mathbf{C} [\mathbf{Q}(\tilde{z}_1) \mathbf{A} \mathbf{Q}(\tilde{z}_2) - \mathbf{\Pi}_1(\tilde{z}_1) \mathbf{S}(\mathbf{A}) \mathbf{\Pi}_2(\tilde{z}_2)] = o_{a.s.}(1). \quad (17)$$

The proof of this lemma is deferred to Appendix B.9.

Remark B.8. Lemma B.7 provides the deterministic equivalent of $\mathbf{Q}(\tilde{z}_1) \mathbf{A} \mathbf{Q}(\tilde{z}_2)$. (Lin & Pan, 2026) established the local laws for the $\mathbf{Q}(\tilde{z}_1) \mathbf{A} \mathbf{Q}(\tilde{z}_2)$. However, their results require $\Re \tilde{z}_1, \Re \tilde{z}_2$ to be sufficiently close to $\text{supp}(\varrho)$ and $\Im \tilde{z}_1, \Im \tilde{z}_2$ to be bounded below by N^{-1+c} , where c is any fixed constant. Lemma B.7 extends the result to other regions.

Remark B.9. To relax the moment assumption, we apply a standard truncation argument commonly used in random matrix theory (e.g., (Yang et al., 2025)). This approach allows us to employ Lemma B.3 under the weaker finite $(8+c)$ -th moment condition, introducing only a negligible additional error term that depends on M but does not affect the leading-order asymptotics of our results. Moreover, a careful examination of the proofs shows that the same moment condition is also sufficient to establish Lemma B.7. Consequently, all of our theoretical conclusions remain valid under this relaxed assumption.

B.2. Proof of Theorem 3.1

To simplify notation, we set $z_1 = -\lambda_t, z_2 = -\lambda_s$. Recalling (2) and (3), we get

$$\begin{aligned} \beta_s - \beta_2 &= \frac{1}{N_2} \mathbf{Q}_2 [\xi \mathbf{X}_2 \mathbf{X}_2^\top \beta_t + (1 - \xi) \mathbf{X}_2 (\mathbf{X}_2^\top \beta_2 + \varepsilon_2)] - \beta_2 \\ &= \frac{1}{N_2} \mathbf{Q}_2 \left[\frac{1}{N_1} \xi \mathbf{X}_2 \mathbf{X}_2^\top \mathbf{Q}_1 \mathbf{X}_1 (\mathbf{X}_1^\top \beta_1 + \varepsilon_1) + (1 - \xi) \mathbf{X}_2 (\mathbf{X}_2^\top \beta_2 + \varepsilon_2) \right] - \beta_2 \\ &= \xi [(\mathbf{I}_M + z_2 \mathbf{Q}_2)(\mathbf{I}_M + z_1 \mathbf{Q}_1) \beta_1 - \beta_2] + \underbrace{\xi \frac{1}{N_1} (\mathbf{I}_M + z_2 \mathbf{Q}_2) \mathbf{Q}_1 \mathbf{X}_1 \varepsilon_1}_{a_5} \\ &\quad + (1 - \xi) [(\mathbf{I}_M + z_2 \mathbf{Q}_2) \beta_2 - \beta_2] + \underbrace{(1 - \xi) \frac{1}{N_2} \mathbf{Q}_2 \mathbf{X}_2 \varepsilon_2}_{a_6} \\ &= \xi \gamma + \underbrace{\xi z_2 \mathbf{Q}_2 \gamma}_{a_1} + \underbrace{\xi z_1 \mathbf{Q}_1 \beta_1}_{a_2} + \underbrace{z_2 \mathbf{Q}_2 \beta_2}_{a_3} + \underbrace{\xi z_1 z_2 \mathbf{Q}_2 \mathbf{Q}_1 \beta_1}_{a_4} + a_5 + a_6. \end{aligned} \quad (18)$$

By this, we decompose $\mathbf{ER}(\beta_s)$ as follows:

$$\begin{aligned}\mathbf{ER}(\beta_s) &= (\beta_s - \beta_2)^\top \Sigma_2 (\beta_s - \beta_2) \\ &= \xi^2 \gamma^\top \Sigma_2 \gamma + 2 \sum_{i=1}^6 b_i + \sum_{i=1}^6 h_{ii} + \sum_{1 \leq i \neq j \leq 6} h_{ij},\end{aligned}\tag{19}$$

where

$$b_i = \xi \gamma^\top \Sigma_2 \mathbf{a}_i, \quad h_{ii} = \|\Sigma_2^{1/2} \mathbf{a}_i\|^2, \quad h_{ij} = \mathbf{a}_i^\top \Sigma_2 \mathbf{a}_j.$$

Moreover, we denote $\widehat{\mathbf{Bias}} = \|\Sigma_2^{1/2}(\xi \gamma + \mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4)\|^2$ and $\widehat{\mathbf{Var}} = h_{55} + h_{66}$. Next, we compute the limit of each term above.

Let $n \in \mathbb{N}^+$. According to the Definition B.2 and the Borel–Cantelli lemma, we have

$$\chi(n) = o_{a.s.}(1) \text{ if } \chi(n) \prec n^{-c}$$

for any constant $c > 0$. By this, the limits of b_1, b_2, b_3, b_4 can be readily obtained using Lemma B.6:

$$\begin{aligned}b_1 &= \xi^2 z_2 \gamma^\top \mathbf{Q}_2 \Sigma_2 \gamma = \xi^2 z_2 \gamma^\top \Pi_2 \Sigma_2 \gamma + o_{a.s.}(1), \\ b_2 &= \xi^2 z_1 \gamma^\top \Sigma_2 \mathbf{Q}_1 \beta_1 = \xi^2 z_1 \gamma^\top \Sigma_2 \Pi_1 \beta_1 + o_{a.s.}(1), \\ b_3 &= \xi z_2 \beta_2^\top \mathbf{Q}_2 \Sigma_2 \gamma = \xi z_2 \beta_2^\top \Pi_2 \Sigma_2 \gamma + o_{a.s.}(1), \\ b_4 &= \xi^2 z_1 z_2 \gamma^\top \Sigma_2 \mathbf{Q}_2 \mathbf{Q}_1 \beta_1 = \xi^2 z_1 z_2 \gamma^\top \Sigma_2 \Pi_2 \Pi_1 \beta_1 + o_{a.s.}(1),\end{aligned}$$

where the last identity is due to

$$\gamma^\top \Sigma_2 [\mathbf{Q}_2 \mathbf{Q}_1 - \Pi_2 \Pi_1] \beta_1 = \gamma^\top \Sigma_2 [(\mathbf{Q}_2 - \Pi_2) \mathbf{Q}_1 + \Pi_2 (\mathbf{Q}_1 - \Pi_1)] \beta_1 = o_{a.s.}(1).$$

We now consider the terms contributing to $\mathbf{Var}$. By Lemma B.4 and the identity

$$\frac{1}{N_i} \mathbf{Q}_i \mathbf{X}_i \mathbf{X}_i^\top = \mathbf{I}_M + z_1 \mathbf{Q}_i, \quad i = 1, 2.\tag{20}$$

we find that

$$\begin{aligned}& \left| h_{55} - \frac{\xi^2 \sigma^2}{N_1^2} \text{Tr}(\mathbf{I}_M + z_2 \mathbf{Q}_2) \Sigma_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) \mathbf{Q}_1 \mathbf{X}_1 \mathbf{X}_1^\top \mathbf{Q}_1 \right| \\ & \prec \frac{1}{M^2} \|(\mathbf{I}_M + z_2 \mathbf{Q}_2) \Sigma_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) (\mathbf{Q}_1 + z_1 \mathbf{Q}_1^2)\|_F \lesssim \frac{1}{\sqrt{M}}.\end{aligned}\tag{21}$$

For any deterministic matrix $\mathbf{C} \in \mathbb{R}^{M \times M}$ satisfying $\|\mathbf{C}\|$ is bounded, and having the spectral decomposition

$$\mathbf{C} = \sum_{i=1}^M \lambda_i \mathbf{u}_i \mathbf{v}_i^\top,$$

we have by Lemma B.6 that

$$\begin{aligned}\frac{1}{M} \text{Tr} \mathbf{C} \mathbf{Q}_1 \mathbf{Q}_2 &= \frac{1}{M} \sum_{i=1}^M \lambda_i \text{Tr} \mathbf{u}_i \mathbf{v}_i^\top \mathbf{Q}_1 \mathbf{Q}_2 \\ &= \frac{1}{M} \sum_{i=1}^M \lambda_i \mathbf{v}_i^\top \mathbf{Q}_1 \mathbf{Q}_2 \mathbf{u}_i \\ &= \frac{1}{M} \sum_{i=1}^M \lambda_i \mathbf{v}_i^\top \Pi_1 \Pi_2 \mathbf{u}_i + o_{a.s.}(1) \\ &= \frac{1}{M} \text{Tr} \mathbf{C} \Pi_1 \Pi_2 + o_{a.s.}(1).\end{aligned}\tag{22}$$

Similarly, by recalling the notation $\Pi_i \mathcal{S}_i(\mathbf{I}_M) \Pi_i = \Pi'_i$ for $i = 1, 2$, one may check by Lemma B.7 that

$$\begin{aligned} \frac{1}{M} \text{Tr} \mathbf{C} \mathbf{Q}_2 \mathbf{Q}_1^2 &= \frac{1}{M} \text{Tr} \mathbf{C} \Pi_2 \Pi'_1 + o_{a.s.}(1), \\ \frac{1}{M} \text{Tr} \mathbf{C} \mathbf{Q}_1 \mathbf{Q}_2^2 &= \frac{1}{M} \text{Tr} \mathbf{C} \Pi_1 \Pi'_2, \\ \frac{1}{M} \text{Tr} \mathbf{Q}_2 \mathbf{C} \mathbf{Q}_2 \mathbf{Q}_1^2 &= \frac{1}{M} \text{Tr} \Pi_2 \mathcal{S}_2(\mathbf{C}) \Pi_2 \Pi'_1 + o_{a.s.}(1), \\ \frac{1}{M} \text{Tr} \mathbf{Q}_1 \mathbf{C} \mathbf{Q}_1 \mathbf{Q}_2^2 &= \frac{1}{M} \text{Tr} \Pi_1 \mathcal{S}_1(\mathbf{C}) \Pi_1 \Pi'_2 + o_{a.s.}(1). \end{aligned} \quad (23)$$

Then by Lemma B.6, Lemma B.7, (20) and (21), for $\xi \neq 0$, we have

$$\begin{aligned} \frac{h_{55}}{\xi^2 \sigma^2} &= \frac{1}{N_1} \text{Tr}(\mathbf{I}_M + z_2 \mathbf{Q}_2) \Sigma_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) (\mathbf{Q}_1 + z_1 \mathbf{Q}_1^2) \\ &= \frac{1}{N_1} \text{Tr} [\Sigma_2 + z_2 \Sigma_2 \mathbf{Q}_2 + z_2 \mathbf{Q}_2 \Sigma_2 + z_2^2 \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_2] [\Pi_1 + z_1 \Pi'_1] + o_{a.s.}(1) \\ &= \frac{1}{N_1} \text{Tr} [\Sigma_2 + 2z_2 \Sigma_2 \Pi_2 + z_2^2 \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2] [\Pi_1 + z_1 \Pi'_1] \\ &\quad + o_{a.s.}(1). \end{aligned}$$

Likewise, we have by Lemma B.6 that

$$\begin{aligned} h_{66} &= (1 - \xi)^2 \sigma^2 \frac{1}{N_2} \text{Tr} \Sigma_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) \mathbf{Q}_2 \\ &= (1 - \xi)^2 \sigma^2 \frac{1}{N_2} \text{Tr} \Sigma_2 (\Pi_2 + z_2 \Pi'_2) + o_{a.s.}(1). \end{aligned} \quad (24)$$

Let $d = \min \{ \text{dist}(z_1, \mathbb{R}^+), \text{dist}(z_2, \mathbb{R}^+) \}$. According to Lemma B.4 and the fact following from (20) that

$$\frac{1}{\sqrt{N_i}} \|\mathbf{Q}_i \mathbf{X}_i\| = \sqrt{\|\mathbf{Q}_i + z_i \mathbf{Q}_i^2\|} \leq \sqrt{d^{-1} + d^{-2} |z_i|} \lesssim 1,$$

one has for $j = 1, 2, 3, 4$,

$$\begin{aligned} |h_{5j}| &= |h_{j5}| \prec \frac{\sigma}{M} \|\mathbf{X}_1^\top \mathbf{Q}_1 (\mathbf{I}_M + z_1 \mathbf{Q}_2) \Sigma_2 \mathbf{a}_j\| \lesssim \frac{1}{\sqrt{M}}, \\ |h_{6j}| &= |h_{j6}| \prec \frac{\sigma}{M} \|\mathbf{X}_2^\top \mathbf{Q}_2 \Sigma_2 \mathbf{a}_j\| \lesssim \frac{1}{\sqrt{M}}, \end{aligned}$$

and

$$|b_5 + b_6| \prec \frac{1}{\sqrt{M}}.$$

Using Lemma B.4 again, it can be shown that

$$\begin{aligned} |h_{65}| &= |h_{56}| \prec \frac{\sigma^2}{M^2} \|\mathbf{X}_2^\top \mathbf{Q}_2 \Sigma_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) \mathbf{Q}_1 \mathbf{X}_1\|_{\text{F}} \\ &\lesssim \frac{\sigma^2}{M} \sqrt{\frac{N_2 \|\mathbf{X}_2^\top \mathbf{Q}_2 \Sigma_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) \mathbf{Q}_1 \mathbf{X}_1\|^2}{M^2}} \lesssim \frac{1}{\sqrt{M}}. \end{aligned}$$

Therefore, we get

$$b_5 + b_6 + h_{65} + h_{56} + \sum_{j=1}^4 (h_{5j} + h_{j5} + h_{6j} + h_{j6}) = o_{a.s.}(1).$$

We now turn to the terms $h_{ii}, i = 1, 2, 3, 4$. By Lemma B.7, we have

$$\begin{aligned}
 h_{11} &= \xi^2 z_2^2 \gamma^\top \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_2 \gamma \\
 &= \xi^2 z_2^2 \gamma^\top \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 \gamma + o_{a.s.}(1), \\
 h_{22} &= \xi^2 z_1^2 \beta_1^\top \mathbf{Q}_1 \Sigma_2 \mathbf{Q}_1 \beta_1 \\
 &= \xi^2 z_1^2 \beta_1^\top \Pi_1 \mathcal{S}_1(\Sigma_2) \Pi_1 \beta_1 + o_{a.s.}(1), \\
 h_{33} &= z_2^2 \beta_2^\top \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_2 \beta_2 \\
 &= z_2^2 \beta_2^\top \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 \beta_2 + o_{a.s.}(1), \\
 h_{44} &= \xi^2 z_1^2 z_2^2 \beta_1^\top \mathbf{Q}_1 \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_2 \mathbf{Q}_1 \beta_1 \\
 &= \xi^2 z_1^2 z_2^2 \mathbb{E} \beta_1^\top \mathbf{Q}_1 \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 \mathbf{Q}_1 \beta_1 + o_{a.s.}(1) \\
 &= \xi^2 z_1^2 z_2^2 \beta_1^\top \Pi_1 \mathcal{S}_1(\Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2) \Pi_1 \beta_1 + o_{a.s.}(1).
 \end{aligned}$$

Similarly, one can obtain the limits of the remaining terms in $h_{ij}, 1 \leq i, j \leq 6$:

$$\begin{aligned}
 h_{12} &= h_{21} = \xi^2 z_1 z_2 \gamma^\top \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_1 \beta_1 \\
 &= \xi^2 z_1 z_2 \gamma^\top \Pi_2 \Sigma_2 \Pi_1 \beta_1 + o_{a.s.}(1), \\
 h_{13} &= h_{31} = \xi z_2^2 \beta_2^\top \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_2 \gamma \\
 &= \xi z_2^2 \beta_2^\top \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 \gamma + o_{a.s.}(1), \\
 h_{14} &= h_{41} = \xi^2 z_1 z_2^2 \gamma^\top \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_2 \mathbf{Q}_1 \beta_1 \\
 &= \xi^2 z_1 z_2^2 \gamma^\top \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 \Pi_1 \beta_1 + o_{a.s.}(1), \\
 h_{23} &= h_{32} = \xi z_1 z_2 \beta_1^\top \mathbf{Q}_1 \Sigma_2 \mathbf{Q}_2 \beta_2 \\
 &= \xi z_1 z_2 \beta_1^\top \Pi_1 \Sigma_2 \Pi_2 \beta_2 + o_{a.s.}(1), \\
 h_{24} &= h_{42} = \xi^2 z_1^2 z_2 \beta_1^\top \mathbf{Q}_1 \Sigma_2 \mathbf{Q}_2 \mathbf{Q}_1 \beta_1 \\
 &= \xi^2 z_1^2 z_2 \beta_1^\top \mathbf{Q}_1 \Sigma_2 \Pi_2 \mathbf{Q}_1 \beta_1 + o_{a.s.}(1) \\
 &= \xi^2 z_1^2 z_2 \beta_1^\top \Pi_1 \mathcal{S}_1(\Sigma_2 \Pi_2) \Pi_1 \beta_1 + o_{a.s.}(1), \\
 h_{34} &= h_{43} = \xi z_1 z_2^2 \beta_2^\top \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_2 \mathbf{Q}_1 \beta_1 \\
 &= \xi z_1 z_2^2 \beta_2^\top \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 \Pi_1 \beta_1 + o_{a.s.}(1).
 \end{aligned}$$

Combining the above estimates, we conclude the proof of Theorem 3.1.

B.3. Proof of Theorem 3.3

We use the same notation as in Appendix B.2. Note that $\gamma = \beta_1 - \beta_2 = 0$. Denoting

$$\mathbf{H} = \xi z_1 \mathbf{Q}_1 + z_2 \mathbf{Q}_2 + \xi z_1 z_2 \mathbf{Q}_2 \mathbf{Q}_1, \quad (25)$$

by (18), we have

$$\beta_s - \beta = \mathbf{H} \beta + \mathbf{a}_5 + \mathbf{a}_6.$$

Hence, the excess risk becomes

$$\begin{aligned}
 \mathbf{ER}(\beta_s) &= \|\Sigma_2^{1/2}(\beta_s - \beta)\|^2 \\
 &= \beta^\top \mathbf{H}^\top \Sigma_2 \mathbf{H} \beta + 2 \sum_{i=5,6} \beta^\top \mathbf{H}^\top \Sigma_2 \mathbf{a}_i + \sum_{i=5,6} h_{ii}.
 \end{aligned}$$

Using Lemma B.4, by Assumption 3.2 we have

$$\beta^\top \mathbf{H}^\top \Sigma_2 \mathbf{H} \beta - \frac{\tilde{\sigma}^2}{M} \text{Tr} \mathbf{H}^\top \Sigma_2 \mathbf{H} \prec \frac{\tilde{\sigma}^2}{M} \|\mathbf{H}^\top \Sigma_2 \mathbf{H}\|_F \lesssim \frac{1}{\sqrt{M}}.$$

By (25), we have

$$\begin{aligned}
 \frac{1}{M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{H} \mathbf{H}^\top &= \frac{1}{M} \left[\xi^2 z_1^2 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_1^2 + \xi z_1 z_2 \text{Tr} \mathbf{\Sigma}_2 [\mathbf{Q}_1 \mathbf{Q}_2 + \mathbf{Q}_2 \mathbf{Q}_1] \right. \\
 &\quad + z_2^2 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2^2 + \xi^2 z_1^2 z_2 \text{Tr} \mathbf{\Sigma}_2 [\mathbf{Q}_2 \mathbf{Q}_1^2 + \mathbf{Q}_1^2 \mathbf{Q}_2] \\
 &\quad \left. + 2\xi z_1 z_2^2 \text{Tr} \mathbf{Q}_2 \mathbf{\Sigma}_2 \mathbf{Q}_2 \mathbf{Q}_1 + \xi^2 z_1^2 z_2^2 \text{Tr} \mathbf{Q}_2 \mathbf{\Sigma}_2 \mathbf{Q}_2 \mathbf{Q}_1^2 \right] \\
 &= \sum_{i=1}^6 t_i,
 \end{aligned} \tag{26}$$

where

$$\begin{aligned}
 t_1 &= \frac{1}{M} \xi^2 z_1^2 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_1^2, t_2 = \frac{2}{M} \xi z_1 z_2 \text{Tr} \mathbf{Q}_1 \mathbf{Q}_2, t_3 = \frac{z_2^2}{M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2^2, \\
 t_4 &= 2 \frac{\xi^2}{M} z_1^2 z_2 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2 \mathbf{Q}_1^2, t_5 = \frac{2\xi z_1 z_2^2}{M} \text{Tr} \mathbf{Q}_2 \mathbf{\Sigma}_2 \mathbf{Q}_2 \mathbf{Q}_1, t_6 = \frac{\xi^2 z_1^2 z_2^2}{M} \text{Tr} \mathbf{Q}_2 \mathbf{\Sigma}_2 \mathbf{Q}_2 \mathbf{Q}_1^2.
 \end{aligned}$$

We next consider the terms $t_i, i = 1, \dots, 6$. In the subsequent proof, we shall make use of Lemma B.6, Lemma B.7 and the property that $\mathbf{\Sigma}_2 \mathbf{\Pi}_2 = \mathbf{\Pi}_2 \mathbf{\Sigma}_2$.

By (22), we have

$$t_2 = \frac{2\xi z_1 z_2}{M} \text{Tr} \mathbf{\Pi}_1 \mathbf{\Pi}_2 \mathbf{\Sigma}_2 + o_{a.s.}(1).$$

$$t_1 = \xi^2 z_1^2 \frac{1}{M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}_1' + o_{a.s.}(1).$$

The limits of t_3, t_4, t_5, t_6 can be derived by (23):

$$t_3 = \frac{z_2^2}{M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}_2' + o_{a.s.}(1),$$

$$t_4 = 2 \frac{\xi^2 z_1^2 z_2}{M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}_2 \mathbf{\Pi}_1' + o_{a.s.}(1),$$

$$t_5 = \frac{2\xi z_1 z_2^2}{M} \text{Tr} \mathbf{\Pi}_2 \mathcal{S}_2(\mathbf{\Sigma}_2) \mathbf{\Pi}_2 \mathbf{\Pi}_1 + o_{a.s.}(1),$$

and

$$t_6 = \frac{\xi^2 z_1^2 z_2^2}{M} \text{Tr} \mathbf{\Pi}_2 \mathcal{S}_2(\mathbf{\Sigma}_2) \mathbf{\Pi}_2 \mathbf{\Pi}_1' + o_{a.s.}(1).$$

Using Lemma B.4, we find

$$|\boldsymbol{\beta}^\top \mathbf{H}^\top \mathbf{\Sigma}_2 \mathbf{a}_5| \prec \frac{1}{M^{3/2}} \|\mathbf{H}^\top \mathbf{\Sigma}_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) \mathbf{Q}_1 \mathbf{X}_1\|_F \lesssim \frac{1}{\sqrt{M}},$$

$$|\boldsymbol{\beta}^\top \mathbf{H}^\top \mathbf{\Sigma}_2 \mathbf{a}_6| \prec \frac{1}{M^{3/2}} \|\mathbf{Q}_2 \mathbf{X}_2\|_F \lesssim \frac{1}{\sqrt{M}}.$$

Therefore, the terms $\boldsymbol{\beta}^\top \mathbf{H}^\top \mathbf{\Sigma}_2 \mathbf{a}_i, i = 5, 6$ are ignorable. The proof is now complete.

B.4. Proof of Corollary 3.8

Letting $\lambda_t = \lambda_s = 0$, by (3), we obtain

$$\begin{aligned}\beta_s &= \xi \beta_1^{\text{OLS}} + (1 - \xi) \beta_2^{\text{OLS}} \\ &= \xi (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{X}_1 (\mathbf{X}_1^\top \beta_1 + \varepsilon_1) + (1 - \xi) (\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \mathbf{X}_2 (\mathbf{X}_2^\top \beta_2 + \varepsilon_2) \\ &= \xi \beta_1 + (1 - \xi) \beta_2 + \xi (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{X}_1 \varepsilon_1 + (1 - \xi) (\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \mathbf{X}_2 \varepsilon_2.\end{aligned}$$

Plugging this into $\mathbf{ER}(\beta_s)$, one may obtain that

$$\begin{aligned}\mathbf{ER}(\beta_s) &= \|\Sigma_2^{1/2}(\beta_2 - \beta_s)\|^2 \\ &= \|\Sigma_2^{1/2}[\xi \gamma + \xi (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{X}_1 \varepsilon_1 + (1 - \xi) (\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \mathbf{X}_2 \varepsilon_2]\|^2 \\ &= \widehat{\mathbf{Bias}} + h_1 + h_2 + 2h_3 + 2h_4 + 2h_5,\end{aligned}$$

where

$$\begin{aligned}h_1 &= \xi^2 \varepsilon_1^\top \mathbf{X}_1^\top (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \Sigma_2 (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{X}_1 \varepsilon_1, \\ h_2 &= (1 - \xi)^2 \varepsilon_2^\top \mathbf{X}_2^\top (\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \Sigma_2 (\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \mathbf{X}_2 \varepsilon_2, \\ h_3 &= \xi^2 \gamma^\top \Sigma_2 (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{X}_1 \varepsilon_1, \\ h_4 &= \xi (1 - \xi) \gamma^\top \Sigma_2 (\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \mathbf{X}_2 \varepsilon_2, \\ h_5 &= \xi (1 - \xi) \varepsilon_1^\top \mathbf{X}_1^\top (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \Sigma_2 (\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \mathbf{X}_2 \varepsilon_2.\end{aligned}$$

By Lemmas B.3-B.4, we have with high probability,

$$\begin{aligned}|h_2 - (1 - \xi)^2 \sigma^2 \text{Tr}(\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \Sigma_2| &= |h_2 - (1 - \xi)^2 \sigma^2 \text{Tr}(\mathbf{Z}_2 \mathbf{Z}_2^\top)^\top| \\ &\prec (1 - \xi)^2 \sigma^2 \|(\mathbf{Z}_2 \mathbf{Z}_2^\top)^{-1}\|_F \\ &= (1 - \xi)^2 \sigma^2 \sqrt{\sum_{i=1}^M \lambda_i^{-2} (\mathbf{Z}_2 \mathbf{Z}_2^\top)} \\ &\lesssim (1 - \xi)^2 \sigma^2 \frac{1}{\sqrt{M}} \text{Tr}(\mathbf{Z}_2 \mathbf{Z}_2^\top)^{-1} \\ &\lesssim \frac{1}{\sqrt{M}}.\end{aligned}\tag{27}$$

Lemma B.6 implies that with high probability,

$$\text{Tr}(\mathbf{Z}_2 \mathbf{Z}_2^\top)^{-1} = \frac{M}{N_2 - M} + o_{a.s.}(1).$$

Combining this with (27), we obtain with high probability,

$$h_2 = (1 - \xi)^2 \sigma^2 \frac{M}{N_2 - M} (1 + o_{a.s.}(1)).$$

Similarly, one may derive with high probability,

$$\begin{aligned}|h_5| &\prec \sigma^2 \|\mathbf{X}_1^\top (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \Sigma_2 (\mathbf{X}_2 \mathbf{X}_2^\top)^{-1} \mathbf{X}_2\|_F \\ &= \sigma^2 \sqrt{\text{Tr}(\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \Sigma_2 (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1}} \lesssim \frac{1}{\sqrt{M}} \widehat{\mathbf{Var}}.\end{aligned}$$

Using Lemmas B.3-B.4, the following estimate holds with high probability,

$$\begin{aligned}|h_1 - \xi^2 \sigma^2 \text{Tr}(\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \Sigma_2| &\prec \|\Sigma_2 (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1}\|_F \\ &\lesssim \sqrt{M \|\Sigma_2\|^2 \|\Sigma_1\|^{-2} \|(\mathbf{Z}_1 \mathbf{Z}_1^\top)^{-1}\|^2} \lesssim \frac{1}{\sqrt{M}}.\end{aligned}$$

Then by Lemma B.3 and Lemma B.6, one has with high probability,

$$\begin{aligned}\text{Tr}(\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{\Sigma}_2 &= \frac{1}{N_1} \text{Tr} \left(\frac{1}{N_1} \mathbf{Z}_1 \mathbf{Z}_1^\top \right)^{-1} \mathbf{\Sigma}_1^{-1/2} \mathbf{\Sigma}_2 \mathbf{\Sigma}_1^{-1/2} \\ &= \frac{1}{N_1} \frac{N_1}{N_1 - M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Sigma}_1^{-1} + o_{a.s.}(1).\end{aligned}$$

Therefore, for $\xi \neq 0$, we get with high probability,

$$\frac{1}{\xi^2 \sigma^2} h_1 = \text{Tr}(\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{\Sigma}_2 + o_{a.s.}(1) = \frac{1}{N_1 - M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Sigma}_1^{-1} + o_{a.s.}(1).$$

We note that

$$\|\gamma\|^2 \gtrsim \widehat{\mathbf{Bias}} = \xi^2 \|\mathbf{\Sigma}_2^{1/2} \gamma\|^2 \gtrsim \lambda_{\min}(\mathbf{\Sigma}_2) \|\gamma\|^2 \gtrsim \|\gamma\|^2.$$

Since $\sigma_M(\mathbf{\Sigma}_1) \lesssim 1$, $\sigma_1(\mathbf{\Sigma}_2) \lesssim 1$, it is easy to see $\widehat{\mathbf{Var}} \sim 1$. Using Lemmas B.3-B.4, we get with high probability

$$\begin{aligned}|h_3| &\prec \xi^2 \sigma \|\gamma^\top \mathbf{\Sigma}_2 (\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{X}_1\| \\ &\leq \xi^2 \sigma \sqrt{\widehat{\mathbf{Bias}}} \|\mathbf{\Sigma}_2^{1/2}\| \|(\mathbf{X}_1 \mathbf{X}_1^\top)^{-1} \mathbf{X}_1\| \\ &\lesssim \frac{\sqrt{\widehat{\mathbf{Bias}}}}{M^{1/4}} \frac{1}{M^{1/4}} \leq \frac{\widehat{\mathbf{Bias}}}{\sqrt{M}} + \frac{1}{\sqrt{M}} \\ &\lesssim \frac{1}{\sqrt{M}} (\widehat{\mathbf{Bias}} + \widehat{\mathbf{Var}}).\end{aligned}$$

Similarly, we can estimate with high probability

$$|h_4| \prec \frac{1}{\sqrt{M}} (\widehat{\mathbf{Bias}} + \widehat{\mathbf{Var}}).$$

Combining the above estimates on $h_i, i = 1, 2, 3, 4, 5$, the proof of Corollary 3.8 is completed.

B.5. Proof of Theorem 3.7

For simplicity, we present the proof only for deterministic β_1 and β_2 ; the extension to the random case follows by similar reasoning and is therefore omitted. Denote $\mathbf{P}_{X_1}$ and $\mathbf{P}_{X_2}$ by

$$\mathbf{P}_{X_1} = (\mathbf{X}_1 \mathbf{X}_1^\top)^+ \mathbf{X}_1 \mathbf{X}_1^\top, \quad \mathbf{P}_{X_2} = (\mathbf{X}_2 \mathbf{X}_2^\top)^+ \mathbf{X}_2 \mathbf{X}_2^\top.$$

Note that for any rectangular matrix $\mathbf{A}$ and compatible $\mathbf{B}$,

$$(\mathbf{A} \mathbf{A}^\top)^+ \mathbf{B} = \lim_{\lambda \rightarrow 0^+} (\mathbf{A} \mathbf{A}^\top + \lambda \mathbf{I}_M)^{-1} \mathbf{B}.$$

We can apply this to $\mathbf{A}_i = \frac{1}{\sqrt{N_i}} \mathbf{X}_i$ for $i = 1, 2$ and rewrite the bias as

$$\mathbf{Bias} = \lim_{\lambda \rightarrow 0^+} f_M(\lambda),$$

where

$$f_M(\lambda) = \xi^2 \gamma^\top \mathbf{\Sigma}_2 \gamma + 2 \sum_{i=1}^6 b_i + \sum_{i=1}^4 h_{ii} + 2 \sum_{1 \leq i \neq j \leq 4} h_{ij},$$

and all terms on the right-hand side are given in Section B.2, under the setting $\lambda_t = \lambda_s = \lambda$. It is straightforward to see that $|f_M(\lambda)| \lesssim 1$. Now we consider $f'_M(\lambda)$. Let $\lambda_{\min}^+(\cdot)$ denote the smallest positive eigenvalue. Lemma B.3 implies that for $i = 1, 2$,

$$\frac{1}{N_i} \lambda_{\max}(\mathbf{X}_i \mathbf{X}_i^\top) \leq 2\sigma_1^i \left(1 + \sqrt{\frac{M}{N_i}}\right)^2, \quad \frac{1}{N_i} \lambda_{\min}^+(\mathbf{X}_i \mathbf{X}_i^\top) \geq \frac{1}{2} \sigma_M^i \left(1 - \sqrt{\frac{M}{N_i}}\right)^2, \quad a.s.$$

Recall that $\|\beta_1\|, \|\beta_2\| \leq c$. Then, by (18), we have with high probability

$$\begin{aligned} \left| \frac{d}{d\lambda} h_{22} \right| &= \left| \frac{d}{d\lambda} \xi^2 \lambda^2 \beta_1^\top \mathbf{Q}_1 \Sigma_2 \mathbf{Q}_1 \beta_1 \right| \\ &= 2\xi^2 \left| \lambda \beta_1^\top \mathbf{Q}_1^2 \frac{1}{N_1} \mathbf{X}_1 \mathbf{X}_1^\top \Sigma_2 \mathbf{Q}_1 \beta_1 \right| \\ &\leq 2\xi^2 \|\beta_1\|^2 \|\lambda \mathbf{Q}_1\| \|\Sigma_2\| \left\| \mathbf{Q}_1^2 \frac{1}{N_1} \mathbf{X}_1 \mathbf{X}_1^\top \right\| \\ &\leq C_\xi \frac{\lambda_{\max}(\mathbf{X}_1 \mathbf{X}_1^\top / N_1)}{(\lambda_{\min}(\mathbf{X}_1 \mathbf{X}_1^\top / N_1) + \lambda)^2} \lesssim 1. \end{aligned}$$

The remaining terms in $f'_M(\lambda)$ can be bounded in a similar manner, and hence $|f'_M(\lambda)| \lesssim 1$ almost surely. Therefore, $\{f_M(x)\}$ is equicontinuous almost surely. By the Arzela-Ascoli theorem, f_M converges uniformly to its limit f , almost surely. By the Moore-Osgood theorem, we can exchange limits and get,

$$\lim_{M \rightarrow \infty} \lim_{\lambda \rightarrow 0^+} f_M(\lambda) = \lim_{\lambda \rightarrow 0^+} \lim_{M \rightarrow \infty} f_M(\lambda) = f(0^+), \quad a.s.$$

Similarly, letting $g_M(\lambda) = h_{55} + h_{66}$ with h_{55} and h_{66} as defined in Section B.2 under the setting $\lambda_t = \lambda_s = \lambda$, we get g_M converges uniformly to its limit, g , and

$$\lim_{M \rightarrow \infty} \lim_{\lambda \rightarrow 0^+} g_M(\lambda) = \lim_{\lambda \rightarrow 0^+} \lim_{M \rightarrow \infty} g_M(\lambda) = g(0^+), \quad a.s.$$

By Theorem 3.1, $f = \widehat{\text{Bias}}$ and $g = \widehat{\text{Var}}$ under the setting $\lambda = \lambda_t = \lambda_s$. It is easy to verify that the remaining terms appearing in 19 are asymptotically negligible, and that f, g are right-continuous. The proof is completed.

B.6. Preliminary analysis of double descent

In this section, we focus on ridgeless regression under the scenario where $\beta_1 = \beta_2 = \beta$ is a random vector with Assumptions 2.1-3.2 being satisfied. We denote

$$\mathbf{Q}_i = \left(\frac{1}{N_i} \mathbf{X}_i \mathbf{X}_i^\top \right)^+, \quad \mathbf{P}_i = \frac{1}{N_i} \mathbf{Q}_i \mathbf{X}_i \mathbf{X}_i^\top.$$

Assuming $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$, we write, by (18),

$$\beta_s - \beta = \underbrace{\xi \mathbf{P}_2 \mathbf{P}_1 \beta + (1 - \xi) \mathbf{P}_2 \beta - \beta}_{\mathbf{b}} + \underbrace{\frac{\xi}{N_1} \mathbf{P}_2 \mathbf{Q}_1 \mathbf{X}_1 \varepsilon_1}_{\mathbf{v}_1} + \underbrace{\frac{(1 - \xi)}{N_2} \mathbf{Q}_2 \mathbf{X}_2 \varepsilon_2}_{\mathbf{v}_2}.$$

Similar to the proof of Theorem 3.1, we find that

$$\mathbf{b}^\top \mathbf{v}_1 + \mathbf{b}^\top \mathbf{v}_2 + \mathbf{v}_1^\top \mathbf{v}_2 = o_{a.s.}(1).$$

Then by Corollary B.4, we have

$$\begin{aligned} \text{ER}(\beta_s) &= \mathbf{b}^\top \mathbf{b} + \sigma^2 \left(\frac{\xi^2}{N_1} \text{Tr} \mathbf{Q}_1 \mathbf{P}_2 \mathbf{P}_1 + \frac{(1 - \xi)^2}{N_2} \text{Tr} \mathbf{Q}_2 \mathbf{P}_2 \right) + o_{a.s.}(1) \\ &= \mathbf{b}^\top \mathbf{b} + \sigma^2 \left(\frac{\xi^2}{N_1} \text{Tr} \mathbf{Q}_1 \mathbf{P}_2 + \frac{(1 - \xi)^2}{N_2} \text{Tr} \mathbf{Q}_2 \right) + o_{a.s.}(1). \end{aligned} \tag{28}$$

We denote $\gamma_1 = \frac{M}{N_1}, \gamma_2 = \frac{M}{N_2}$. We partition (γ_1, γ_2) into four disjoint regions to analyze the asymptotic behavior of the excess risk.

(1) Case 1: $\gamma_1, \gamma_2 \leq 1 - \tau$. Recalling Corollary 3.8, we have

$$\text{ER}(\beta_s) = (1 - \xi)^2 \sigma^2 \frac{\gamma_2}{1 - \gamma_2} + \xi^2 \sigma^2 \frac{\gamma_1}{1 - \gamma_1} + o_{a.s.}(1). \tag{29}$$

(2) Case 2: $\gamma_1 \geq 1 + \tau, \gamma_2 \leq 1 - \tau$. In this case, we have

$$\begin{aligned} \mathbf{ER}(\beta_s) &= \xi^2 \frac{\tilde{\sigma}^2}{M} \text{Tr}(\mathbf{I}_M - \mathbf{P}_1) + \sigma^2 \left(\frac{\xi^2}{N_1} \text{Tr} \mathbf{Q}_1 + \frac{(1-\xi)^2}{N_2} \text{Tr} \mathbf{Q}_2 \right) + o_{a.s.}(1) \\ &= \xi^2 \tilde{\sigma}^2 (1 - \gamma_1^{-1}) + \sigma^2 \left(\xi^2 \frac{1}{\gamma_1 - 1} + (1-\xi)^2 \frac{\gamma_2}{1 - \gamma_2} \right) + o_{a.s.}(1). \end{aligned} \quad (30)$$

(3) Case 3: $\gamma_1 \leq 1 - \tau, \gamma_2 \geq 1 + \tau$. By Theorem 3.7, we can obtain that

$$\begin{aligned} \mathbf{ER}(\beta_s) &= \frac{\tilde{\sigma}^2}{M} \text{Tr}(\mathbf{I}_M - \mathbf{P}_2) + \sigma^2 \left(\frac{\xi^2}{N_1} \text{Tr} \mathbf{Q}_1 \mathbf{P}_2 + \frac{(1-\xi)^2}{N_2} \text{Tr} \mathbf{Q}_2 \right) + o_{a.s.}(1) \\ &= \tilde{\sigma}^2 (1 - \gamma_2^{-1}) + \sigma^2 \left(\xi^2 \frac{\gamma_1}{\gamma_2(1 - \gamma_1)} + (1-\xi)^2 \frac{1}{\gamma_2 - 1} \right) + o_{a.s.}(1). \end{aligned} \quad (31)$$

(4) Case 4: $\gamma_1, \gamma_2 \geq 1 + \tau$. Using Theorem 3.7 again, we have

$$\begin{aligned} \mathbf{ER}(\beta_s) &= \frac{\tilde{\sigma}^2}{M} \text{Tr}(-\xi^2 \mathbf{P}_2 \mathbf{P}_1 + (\xi^2 - 1) \mathbf{P}_2 + \mathbf{I}_M) \\ &\quad + \sigma^2 \left(\frac{\xi^2}{N_1} \text{Tr} \mathbf{Q}_1 \mathbf{P}_2 + \frac{(1-\xi)^2}{N_2} \text{Tr} \mathbf{Q}_2 \right) + o_{a.s.}(1) \\ &= \tilde{\sigma}^2 (\xi^2 \gamma_2^{-1} (1 - \gamma_1^{-1}) - \gamma_2^{-1} + 1) + \sigma^2 \left(\xi^2 \frac{1}{\gamma_2(\gamma_1 - 1)} + (1-\xi)^2 \frac{1}{\gamma_2 - 1} \right) \\ &\quad + o_{a.s.}(1). \end{aligned} \quad (32)$$

The above results characterize the asymptotic excess risk in terms of its bivariate dependence on γ_1 and γ_2 . In fact, these findings extend readily to the isotropic setting where $\Sigma_1 = c_1 \mathbf{I}_M, \Sigma_2 = c_2 \mathbf{I}_M$ for some constants $c_1, c_2 > 0$; the details of which are omitted for brevity. Furthermore, for a fixed ξ , maintaining either γ_1 or γ_2 constant allows for a direct observation of the risk's non-monotonic behavior relative to the other parameter, as illustrated in Figures 5 and 6. A complete visualization of the risk surface as a function of both γ_1 and γ_2 is provided in Figure 2.

Figure 7 displays the contour plots of the asymptotic excess risk across four distinct values of ξ . In the anti-learning regime (Figure 7(a)), the risk surface exhibits high sensitivity to γ_2 , as evidenced by the densely packed contours along that axis. However, as ξ increases, this sensitivity gradually shifts toward γ_1 . Intuitively, this reflects the role of the imitation parameter ξ in modulating the influence of the teacher's supervision: a larger ξ amplifies the impact of γ_1 on the overall risk profile. The numerical results in Figures 5–7 are consistent with the theoretical characterizations provided in (29)–(32).

B.7. Proof of Corollary 3.5

The Stieltjes transform of the Marchenko-Pastur distribution is given by

$$\underline{m}_i(z) = \int \frac{d\varrho_{\text{MP},i}(x)}{x - z} = \frac{\left(1 - \frac{M}{N_i}\right) - z - \sqrt{\left(z - 1 - \frac{M}{N_i}\right)^2 - 4\frac{M}{N_i}}}{2\frac{M}{N_i}z}. \quad (33)$$

We take the derivative of $\mathbf{ER}(\beta_s)$ with respect to ξ , and evaluate it at $\xi = 0$:

$$\left. \frac{\partial}{\partial \xi} \mathbf{ER}(\beta_s) \right|_{\xi=0} = \sigma^2 \left(\lambda_t \lambda_s \underline{m}_1 \text{SNR} - \frac{M}{N_2} \right) \frac{d}{dz} (z \underline{m}_2(z)) \Big|_{z=-\lambda_s} + o_{a.s.}(1).$$

Since

$$z \underline{m}_2(z) = \int \frac{z}{x - z} d\varrho_{\text{MP},2}(x) = -1 + \int \frac{x}{x - z} d\varrho_{\text{MP},2}(x),$$

we hence get that

$$\frac{d}{dz} (z \underline{m}_2(z)) \Big|_{z=-\lambda_s} > 0. \quad (34)$$

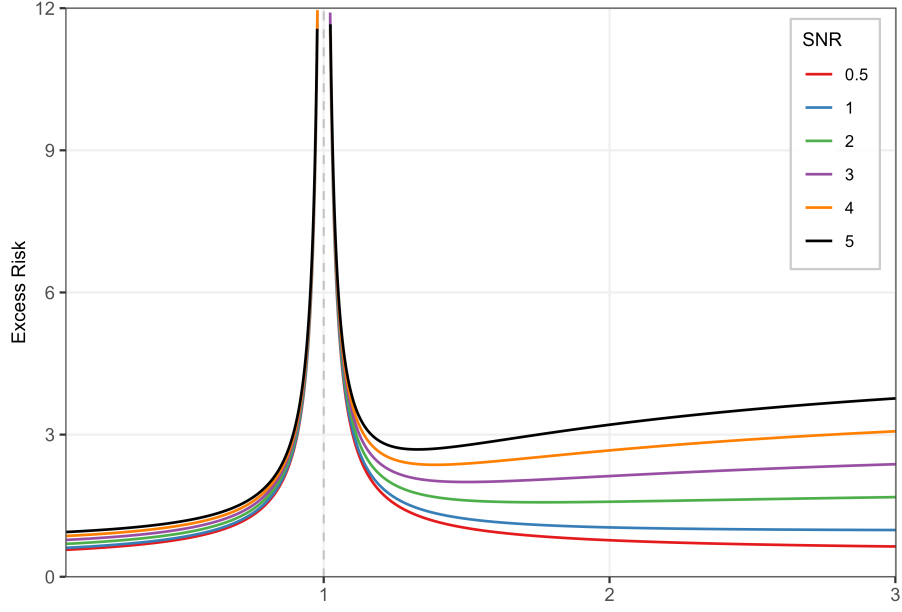


Figure 5. Asymptotic excess risk as a function of γ_2 for the ridgeless regression when SNR varies from 0.5 to 5. Settings: $\sigma^2 = 1$, $\xi = 0.5$, $\gamma_1 = 1.5$, $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$.

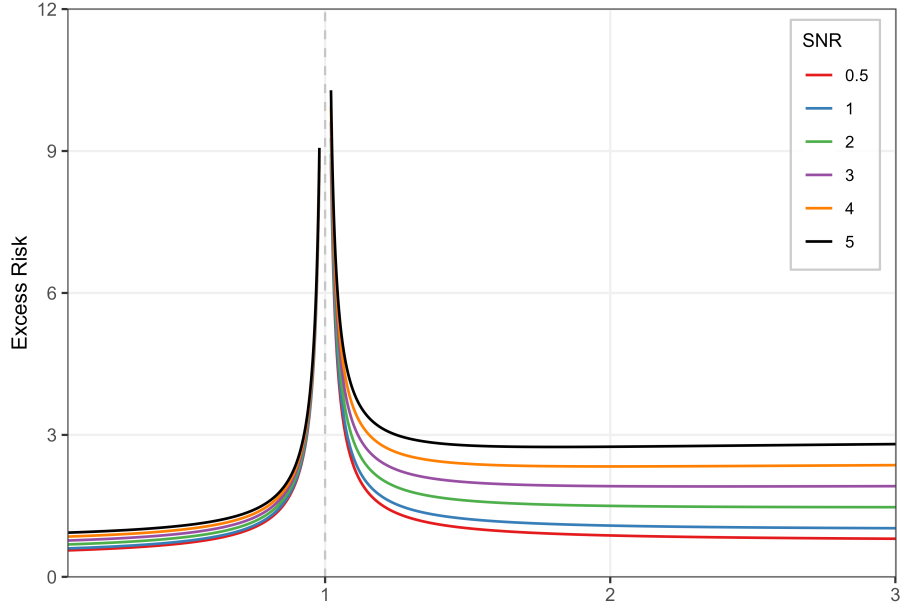


Figure 6. Asymptotic excess risk as a function of γ_1 for the ridgeless regression when SNR varies from 0.5 to 5. Settings: $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$, $\sigma^2 = 1$, $\xi = 0.5$, $\gamma_2 = 1.5$.

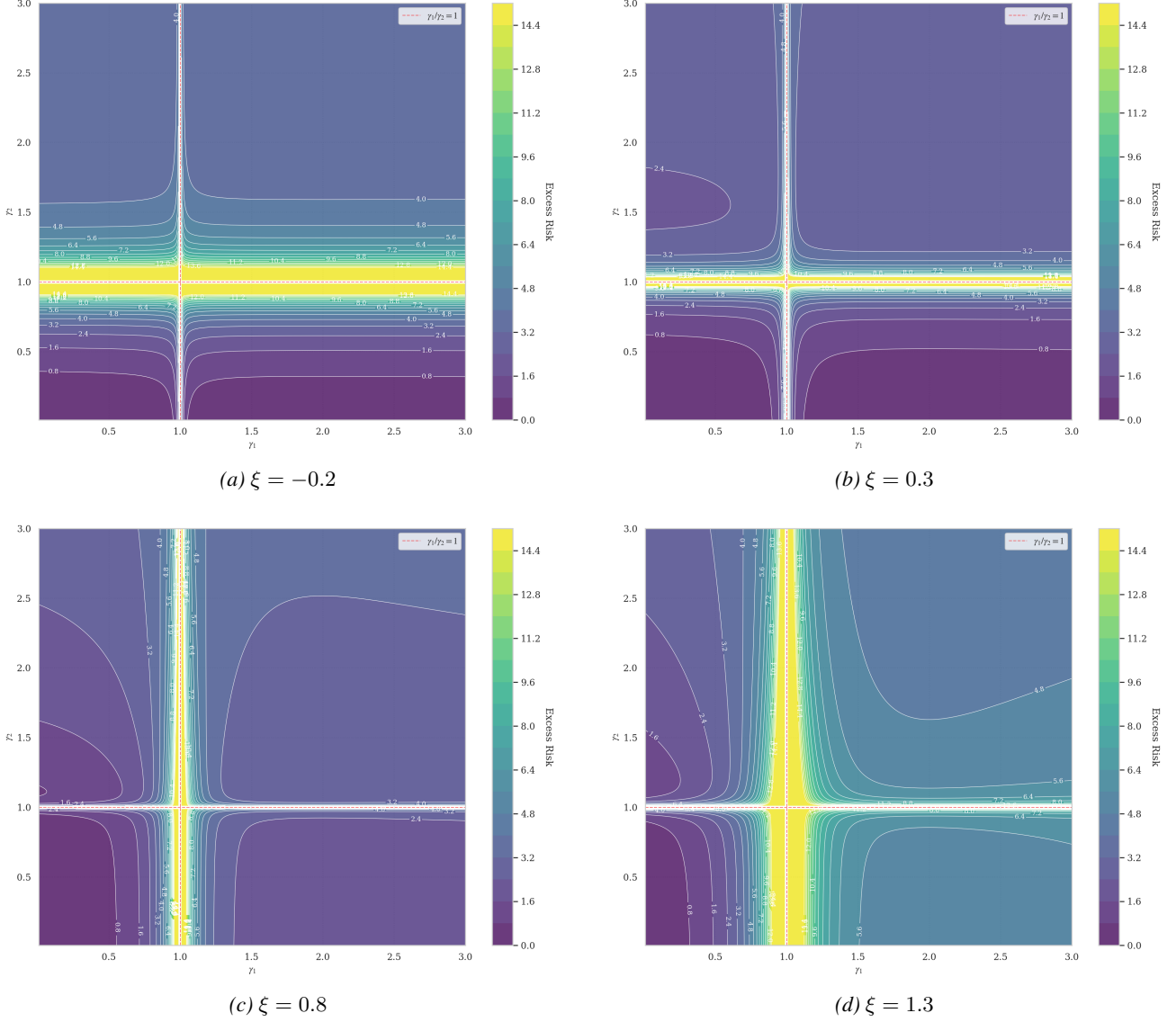


Figure 7. Contour plots of the asymptotic excess risk in ridgeless regression for varying values of ξ . Settings: $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$, $\sigma^2 = 1$, SNR=2.

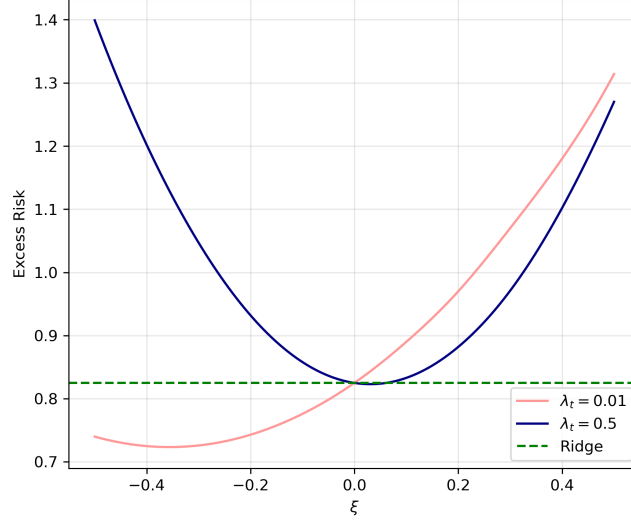


Figure 8. Theoretical excess risk for different λ_t . Settings: $(M, N_1, N_2) = (200, 200, 600)$, $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$, $\lambda_s = 0.5$, $\text{SNR}=4$, $\beta_1 = \beta_2 \sim \mathcal{N}(0, \frac{4}{M})$, $\sigma^2 = 1$.

Therefore, $\frac{\partial}{\partial \xi} \mathbf{ER}(\beta_s)|_{\xi=0}$ and $\lambda_t \lambda_s \text{SNR} - \frac{M}{N_2}$ share the same sign almost surely. That is, the limiting optimal value of ξ is negative when (10) holds, which establishes Corollary 3.5.

We provide an example in Figure 8 to illustrate this corollary. Specifically, when $\lambda_t = 0.5$, the limiting optimal value of ξ is positive, whereas when $\lambda_t = 0.01$, it becomes negative.

B.8. Details for Proposition 3.9 and Remark 3.10

Recall that $\mathbf{ER}_0$ denotes the excess risk of the ridge regression model trained solely on the target domain data.

Assumption B.10. When β_1, β_2 are deterministic, we assume that :

$$\left| \beta_1^\top \lambda_t \lambda_s [\Pi_1 \Sigma_2 \Pi_2 - \lambda_s \Pi_1 \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2] \beta_2 - \frac{\sigma^2}{N_2} \text{Tr}[\Sigma_2(\Pi_2 - \lambda_s \Pi_2')] \right. \\ \left. + \beta_2^\top [\lambda_s^2 \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 - \lambda_s \Pi_2 \Sigma_2] \gamma \right| > c,$$

where c is a positive constant.

Assumption B.11. If $\beta = \beta_1 = \beta_2$ is random, we assume that

$$\left| \frac{\tilde{\sigma}^2}{M} \left[\lambda_t \lambda_s \text{Tr} \Pi_1 \Pi_2 \Sigma_2 - \lambda_t \lambda_s^2 \text{Tr} \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 \Pi_1 \right] - \frac{\sigma^2}{N_2} \text{Tr}[\Sigma_2(\Pi_2 - \lambda_s \Pi_2')] \right| > c,$$

where c is a positive constant.

Proof of Proposition 3.9: (i) Suppose the conditions in Theorem 3.1 hold. Note that

$$\frac{\partial}{\partial \xi} \mathbf{ER}(\beta_s)|_{\xi=0} = 2\beta_1^\top [\lambda_t \lambda_s \Pi_1 \Sigma_2 \Pi_2 - \lambda_t \lambda_s^2 \Pi_1 \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2] \beta_2 \\ + 2\beta_2^\top [\lambda_s^2 \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 - \lambda_s \Pi_2 \Sigma_2] \gamma - \frac{2\sigma^2}{N_2} \text{Tr}[\Sigma_2(\Pi_2 - \lambda_s \Pi_2')] + o_{a.s.}(1).$$

Under the conditions in Theorem 3.1 and Assumption B.10, the asymptotic excess risk is a quadratic function whose minimizer is bounded away from 0. Therefore, $\min_{\xi} \mathbf{ER}(\beta_s)$ is strictly less than $\mathbf{ER}_0$ almost surely.

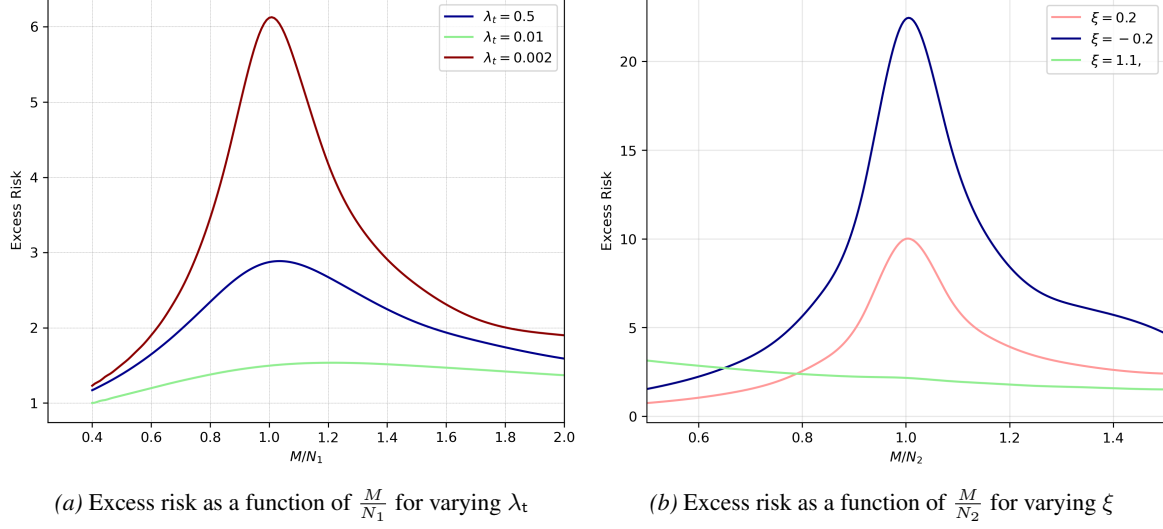


Figure 9. Non-monotone excess risk curves. We set $\Sigma_2 = \mathbf{I}_M$, $\Sigma_1 = \text{diag}(d_1, \dots, d_M)$ where $d_i = 0.64\delta(i \leq \frac{M}{2}) + 0.25\delta(\frac{M}{2} < i \leq M)$. (a) Results are shown for fixed $M = 600$ and $\lambda_s = 0.05$ with different N_1 . (b) Results are shown for fixed $N_2 = 600$ and $(\lambda_t, \lambda_s) = (0.05, 0.001)$, with varying M .

(ii) Similarly, suppose Assumption B.11, under the conditions of Theorem 3.3, the inequality (13) holds by noticing that

$$\begin{aligned} \frac{\partial}{\partial \xi} \mathbf{ER}(\beta_s) \Big|_{\xi=0} &= \frac{2\tilde{\sigma}^2}{M} \left[\lambda_t \lambda_s \text{Tr} \Pi_1 \Pi_2 \Sigma_2 - \lambda_t \lambda_s^2 \text{Tr} \Pi_2 \mathcal{S}_2(\Sigma_2) \Pi_2 \Pi_1 \right] \\ &\quad - \frac{2\sigma^2}{N_2} \text{Tr}[\Sigma_2(\Pi_2 - \lambda_s \Pi_2')] + o_{a.s.}(1). \end{aligned}$$

Further discussion on Remark 3.10: To clarify the dependence of Assumption B.10 on the geometry of $\Sigma_1, \Sigma_2, \beta_1, \beta_2$ and the noise strength σ^2 , we consider a simple example in which $\Sigma_2 = \mathbf{I}_M$. Then we have

$$\begin{aligned} \frac{\partial}{\partial \xi} \mathbf{ER}(\beta_s) \Big|_{\xi=0} &= \lambda_s \lambda_t (\underline{m}_2 - \lambda_s \underline{m}_2') \beta_1^\top \Pi_1 \beta_2 \\ &\quad - \frac{\sigma^2 M}{N_2} (\underline{m}_2 - \lambda_s \underline{m}_2') - \lambda_s (\underline{m}_2 - \lambda_s \underline{m}_2') \beta_2^\top \gamma + o_{a.s.}(1) \\ &= \underbrace{(\lambda_s \lambda_t \beta_1^\top \Pi_1 \beta_2 - \frac{\sigma^2 M}{N_2} - \lambda_s \beta_2^\top \gamma)}_e \frac{d}{dz} (z \underline{m}_2(z)) \Big|_{z=-\lambda_s} + o_{a.s.}(1), \end{aligned} \tag{35}$$

where $\underline{m}_2(z)$ is defined in (33). Recalling (34), we have $|\frac{\partial}{\partial \xi} \mathbf{ER}(\beta_s) \Big|_{\xi=0}| > c$ if $|e| > C$ for some constant C . Below, we discuss two cases, when $\Sigma_1 = \mathbf{I}_M$ and when $\Sigma_1 \neq \mathbf{I}_M$:

- $\Sigma_1 = \mathbf{I}_M$. The term e becomes

$$e = \lambda_s \lambda_t \underline{m}_1 \beta_1^\top \beta_2 - \frac{\sigma^2 M}{N_2} - \lambda_s \beta_2^\top (\beta_1 - \beta_2).$$

Recall that the asymptotic ridge risk is minimized at $\lambda_s^* = \frac{\sigma^2 M}{N_2 \|\beta_2\|^2}$, with asymptotic excess risk $\sigma^2 \frac{M}{N_2} \underline{m}_2(-\lambda_s^*)$ (Hastie et al., 2022). Taking $\lambda_s = \lambda_s^*$, we have

$$e = \lambda_s^* (\lambda_t \underline{m}_1 - 1) \beta_1^\top \beta_2.$$

Note that

$$\lambda_t \underline{m}_1 - 1 = - \int \frac{x}{x + \lambda_t} d\varrho_{\text{MP},1}(x) < 0.$$

Therefore, in a small neighborhood containing 0, $\mathbf{ER}(\beta_s)$ is monotonic in ξ , indicating that the teacher's supervision is helpful – even outperforming the optimal ridge regression – provided that β_1 and β_2 are not asymptotically orthogonal.

- $\Sigma_1 \neq \mathbf{I}_M$. By taking $\lambda_s = \lambda_s^*$, e becomes

$$e = \lambda_s^* (\lambda_t \beta_1^\top \Pi_1 \beta_2 - \beta_2^\top \beta_1) = -\lambda_s^* \sum_{i=1}^M \frac{m_1 \sigma_i}{1 + m_1 \sigma_i} \beta_1^\top \mathbf{u}_i \mathbf{u}_i^\top \beta_2, \quad (36)$$

where m_1 is determined by Lemma 2.3 and $\Sigma_1 = \sum_{i=1}^M \sigma_i \mathbf{u}_i \mathbf{u}_i^\top$ represents the spectral decomposition of Σ_1 . By (36), the alignment of β_i ($i = 1, 2$) with the eigenvectors of Σ_1 determines whether Assumption B.10 is satisfied. Therefore, given $\lambda_s = \lambda_s^*$, under the “help” of covariate shift, even if $\beta_1^\top \beta_2 = 0$, it may still be possible to find a ξ such that $\mathbf{ER}(\beta_s) < \mathbf{ER}_0$, a.s. By comparing with the case where $\Sigma_1 = \mathbf{I}_M$, we find that the presence of covariate shift can, in some cases, be beneficial.

Note that, up to asymptotically negligible terms, $\mathbf{ER}$ can be expressed as a quadratic function of ξ : $\mathbf{ER}(\beta_s, \xi) = A\xi^2 + B\xi + C$. Below we provide closed-form expressions for the asymptotic optimal $\xi^* = -\frac{B}{2A}$ under several common settings.

- (1) When $\gamma = \beta_1 - \beta_2$, $\frac{M}{N_1}, \frac{M}{N_2} < 1 - \tau$,

$$\xi^* = \left(\gamma^\top \Sigma_2 \gamma + \sigma^2 \frac{M}{N_2 - M} + \frac{\sigma^2}{N_1 - M} \text{Tr} \Sigma_2 \Sigma_1^{-1} \right)^{-1} \frac{\sigma^2 M}{N_2 - M} \in (0, 1).$$

- (2) When $\beta = \beta_1 = \beta_2$ is random, and $\Sigma_1 = \Sigma_2 = \mathbf{I}_M$,

$$\xi^* = \frac{\frac{M}{N_2} (\underline{m}_2 - \lambda_s \underline{m}'_2) - \frac{\tilde{\sigma}^2}{\sigma^2} \lambda_t \lambda_s (\underline{m}_1 \underline{m}_2 - \lambda_s \underline{m}'_2 \underline{m}_1)}{A_1 + A_2 + A_3},$$

where

$$\begin{aligned} A_1 &= \frac{\tilde{\sigma}^2}{\sigma^2} (\lambda_t^2 \underline{m}'_1 - 2\lambda_t^2 \lambda_s \underline{m}_2 \underline{m}'_1 + \lambda_t^2 \lambda_s^2 \underline{m}'_1 \underline{m}'_2), \\ A_2 &= \frac{M}{N_1} (\underline{m}_1 - 2\lambda_s \underline{m}_1 \underline{m}_2 + \lambda_s^2 \underline{m}_1 \underline{m}'_2 - \lambda_t \underline{m}'_1 + 2\lambda_t \lambda_s \underline{m}_2 \underline{m}'_1 - \lambda_t \lambda_s^2 \underline{m}'_1 \underline{m}'_2), \\ A_3 &= \frac{M}{N_2} (\underline{m}_2 - \lambda_s \underline{m}'_2). \end{aligned}$$

- (3) When $\beta = \beta_1 = \beta_2$, $\Sigma_2 = \mathbf{I}_M$,

$$\xi^* = \frac{\beta^\top [\lambda_t \lambda_s^2 \underline{m}'_2 \Pi_1 - \lambda_t \lambda_s \underline{m}_2 \Pi_1] \beta + \frac{\sigma^2 M}{N_2} (\underline{m}'_2 - \lambda_s \underline{m}'_2)}{\beta^\top \lambda_t^2 [1 + \lambda_s^2 \underline{m}'_2 - 2\lambda_s \underline{m}_2] \Pi_1' \beta + \frac{\sigma^2 (1 - 2\lambda_s \underline{m}_2 + \lambda_s^2 \underline{m}'_2)}{N_1} \text{Tr} [\Pi_1 - \lambda_t \Pi_1'] + \frac{\sigma^2 M}{N_2} (\underline{m}_2 - \lambda_s \underline{m}'_2)}.$$

B.9. Proof of Lemma B.7

The following result, which is an immediate consequence of Lemma 2.3, will be used in the proof below:

$$-zm = \left(1 + \frac{1}{N} \text{Tr} \Sigma \Pi(z) \right)^{-1}. \quad (37)$$

We abuse notation by writing z_1 and z_2 for $\tilde{z}_1$ and $\tilde{z}_2$, respectively, whenever there is no risk of ambiguity. Without loss of generality, we assume $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$ and z_1, z_2 lie on the negative real axis, as the other cases follow by analogous arguments.

Using standard techniques of martingale decomposition (see, e.g., (Bai & Silverstein, 2010)), we can prove the almost sure convergence of the random part:

$$\mathbf{u}^\top \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{v} = \mathbf{u}^\top \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{v} + o_{a.s.}(1). \quad (38)$$

Therefore, it suffices to consider the term $\mathbf{u}^\top \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{v}$. Let $\sigma_1 \geq \dots \geq \sigma_M$ denote the eigenvalues of Σ . For the sequence of deterministic matrices, we denote $\mathbf{A}_M = o(1)$ if $\|\mathbf{A}_M\| \rightarrow 0$. Since

$$\mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) = \mathbf{Q}(z_1) \mathbf{A} \Pi(z_2) + \mathbf{Q}(z_1) \mathbf{A} (\mathbf{Q}(z_2) - \Pi(z_2)), \quad (39)$$

we obtain by Lemma B.6 that

$$\begin{aligned} \mathbf{u}^\top \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{v} &= \mathbf{u}^\top \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \Pi(z_2) \mathbf{v} + \mathbf{u}^\top \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} (\mathbf{Q}(z_2) - \Pi(z_2)) \mathbf{v} \\ &= \mathbf{u}^\top \Pi(z_1) \mathbf{A} \Pi(z_2) \mathbf{v} + \mathbf{u}^\top \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} (\mathbf{Q}(z_2) - \Pi(z_2)) \mathbf{v} + o(1), \end{aligned} \quad (40)$$

where the second identity follows from Lemma B.6, the Dominated Convergence Theorem and the fact that

$$\|\Pi(z_1)\| = \max_i |z_1 + z_1 m(z_1) \sigma_i|^{-1} \leq |z_1|^{-1}, \quad \|\mathbf{A} \Pi(z_2) \mathbf{v}\| \leq \|\mathbf{A}\| \|\Pi(z_2)\| \leq |z_2|^{-1} \|\mathbf{A}\|.$$

Therefore, our task reduces to finding the deterministic equivalent of

$$\mathbb{E} \mathbf{Q}(z_1) \mathbf{A} (\mathbf{Q}(z_2) - \Pi(z_2)).$$

Denote

$$\mathbf{X}_{-k} = \mathbf{X} - \mathbf{x}_k \mathbf{e}_k^\top, \quad \mathbf{Q}_{-k}(z) = \left(\frac{\mathbf{X}_{-k} \mathbf{X}_{-k}^\top}{N} - z \mathbf{I}_M \right)^{-1}.$$

By Sherman-Morrison formula, one may easily check that

$$\begin{aligned} \mathbf{Q}(z) &= \mathbf{Q}_{-k}(z) - \frac{\frac{1}{N} \mathbf{Q}_{-k}(z) \mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-k}(z)}{1 + \frac{1}{N} \mathbf{x}_k^\top \mathbf{Q}_{-k}(z) \mathbf{x}_k}, \\ \mathbf{Q}(z) \mathbf{x}_k &= \frac{\mathbf{Q}_{-k}(z) \mathbf{x}_k}{1 + \frac{1}{N} \mathbf{x}_k^\top \mathbf{Q}_{-k}(z) \mathbf{x}_k}. \end{aligned} \quad (41)$$

We show here the following result for future use:

$$\frac{1}{N} \mathbb{E} \text{Tr} \mathbf{C} \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) = \frac{1}{N} \mathbb{E} \text{Tr} \mathbf{C} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) + o(1), \quad (42)$$

where $\mathbf{C} \in \mathbb{R}^{M \times M}$ is a deterministic matrix with $\|\mathbf{C}\| \leq C$ for some constant C . We decompose

$$\begin{aligned} &\mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) - \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \\ &= [\mathbf{Q}(z_1) - \mathbf{Q}_{-1}(z_1)] \mathbf{A} \mathbf{Q}(z_2) + \mathbf{Q}_{-1}(z_1) \mathbf{A} [\mathbf{Q}_{-1}(z_2) - \mathbf{Q}_{-1}(z_2)]. \end{aligned}$$

Applying the identity

$$\mathbf{A}^{-1} - \mathbf{B}^{-1} = \mathbf{B}^{-1} (\mathbf{B} - \mathbf{A}) \mathbf{A}^{-1}, \quad (43)$$

we have for $i = 1, 2$, and $\tilde{\mathbf{C}} \in \mathbb{R}^{M \times M}$ with finite spectral norm (where $\tilde{\mathbf{C}}$ may be a deterministic matrix, or a random matrix that is either dependent on or independent of $\mathbf{X}$),

$$\frac{1}{N} |\text{Tr} [\mathbf{Q}(z_i) - \mathbf{Q}_{-1}(z_i)] \tilde{\mathbf{C}}| = \frac{1}{N^2} |\mathbf{x}_1^\top \mathbf{Q}(z_i) \tilde{\mathbf{C}} \mathbf{Q}_{-1}(z_1) \mathbf{x}_1| \leq \frac{C}{N^2} \|\mathbf{x}_1\|^2 = o_{a.s.}(1).$$

We denote $d = \min\{\text{dist}(z_1, \mathbb{R}^+), \text{dist}(z_2, \mathbb{R}^+)\}$. One may easily check that $d \sim 1$. Then by

$$\frac{1}{N} |\text{Tr} [\mathbf{Q}(z_i) - \mathbf{Q}_{-1}(z_i)] \tilde{\mathbf{C}}| \leq \frac{M}{N} (\|\mathbf{Q}(z_i) \tilde{\mathbf{C}}\| + \|\mathbf{Q}_{-1}(z_i) \tilde{\mathbf{C}}\|) \leq \frac{2M}{dN}, \quad \text{for } i = 1, 2,$$

and the Dominated Convergence Theorem, we obtain (42). By similar arguments, we get for any deterministic unit vectors $\mathbf{u}, \mathbf{v}$,

$$\begin{aligned} \mathbf{u}^\top \mathbb{E} \tilde{\mathbf{C}} \mathbf{Q}(z_i) \mathbf{C} \mathbf{v} &= \mathbf{u}^\top \mathbb{E} \tilde{\mathbf{C}} \mathbf{Q}_{-k}(z_i) \mathbf{C} \mathbf{v} + o(1) \\ &= \mathbf{u}^\top \tilde{\mathbf{C}} \boldsymbol{\Pi}(z_i) \mathbf{C} \mathbf{v} + o(1), \quad i = 1, 2, \\ \mathbf{u}^\top \mathbb{E} \tilde{\mathbf{C}} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{C} \mathbf{v} &= \mathbf{u}^\top \mathbb{E} \tilde{\mathbf{C}} \mathbf{Q}_{-k}(z_1) \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{C} \mathbf{v} + o(1), \end{aligned} \quad (44)$$

where $\tilde{\mathbf{C}}$ and $\mathbf{C}$ are deterministic $M \times M$ matrices with finite spectral norms.

We denote

$$\begin{aligned} b_k &= \frac{1}{N} \mathbf{x}_k^\top \mathbf{Q}_{-k}(z_2) \mathbf{x}_k, \quad \tilde{b} = \frac{1}{N} \mathbb{E} \mathbf{x}_k^\top \mathbf{Q}_{-k}(z_2) \mathbf{x}_k, \\ b_k &= \frac{1}{N} \mathbf{x}_k^\top \mathbf{Q}_{-k}(z_1) \mathbf{x}_k, \quad \tilde{b} = \frac{1}{N} \mathbb{E} \mathbf{x}_k^\top \mathbf{Q}_{-k}(z_1) \mathbf{x}_k. \end{aligned}$$

It follows directly from the proof of (42) that

$$\begin{aligned} \tilde{b} &= \frac{1}{N} \mathbb{E} \text{Tr} \boldsymbol{\Sigma} \mathbf{Q}_{-1}(z_2) = \frac{1}{N} \text{Tr} \boldsymbol{\Sigma} \boldsymbol{\Pi}(z_2) + o(1), \\ \tilde{b} &= \frac{1}{N} \mathbb{E} \text{Tr} \boldsymbol{\Sigma} \mathbf{Q}_{-1}(z_2) = \frac{1}{N} \text{Tr} \boldsymbol{\Sigma} \boldsymbol{\Pi}(z_2) + o(1). \end{aligned} \quad (45)$$

Recalling (37), we rewrite $\mathbb{E} \mathbf{Q}(z_1) \mathbf{A} (\mathbf{Q}(z_2) - \boldsymbol{\Pi}(z_2))$ as

$$\begin{aligned} \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} (\mathbf{Q}(z_2) - \boldsymbol{\Pi}(z_2)) &= \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) (\mathbf{I}_M - \mathbf{Q}^{-1}(z_2) \boldsymbol{\Pi}(z_2)) \\ &= \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) (\boldsymbol{\Pi}^{-1}(z_2) - \mathbf{Q}^{-1}(z_2)) \boldsymbol{\Pi}(z_2) \\ &= \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \left(-\frac{1}{N} \mathbf{X} \mathbf{X}^\top - z_2 m \boldsymbol{\Sigma} \right) \boldsymbol{\Pi}(z_2) \\ &= \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \frac{\boldsymbol{\Sigma} \boldsymbol{\Pi}(z_2)}{1 + \frac{1}{N} \text{Tr} \boldsymbol{\Sigma} \boldsymbol{\Pi}(z_2)} - \frac{1}{N} \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{X} \mathbf{X}^\top \boldsymbol{\Pi}(z_2). \end{aligned} \quad (46)$$

An application of (41) yields that

$$\begin{aligned} \frac{1}{N} \sum_{k=1}^N \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{x}_k \mathbf{x}_k^\top &= \frac{1}{N} \sum_{k=1}^N \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \frac{\mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top}{1 + b_k} \\ &= \frac{1}{N(1 + \tilde{b})} \sum_{k=1}^N \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top \left[1 + \frac{\tilde{b} - b_k}{(1 + b_k)} \right] \\ &= \frac{1}{N(1 + \tilde{b})} \left[\sum_{k=1}^N \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top + \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{X} \mathbf{B} \mathbf{X}^\top \right] \\ &= \frac{1}{1 + \tilde{b}} (\mathbb{E} \mathbf{F}_1 + \mathbb{E} \mathbf{F}_2), \end{aligned} \quad (47)$$

where $\mathbf{B} = \text{diag}(\tilde{b} - b_1, \dots, \tilde{b} - b_N)$, and

$$\mathbf{F}_1 = \frac{1}{N} \sum_{k=1}^N \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top, \quad \mathbf{F}_2 = \frac{1}{N} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{X} \mathbf{B} \mathbf{X}^\top.$$

We now bound the spectral norm of $\mathbf{F}_2$. Define the event

$$\mathcal{E} = \left\{ \frac{1}{N} \|\mathbf{Z} \mathbf{Z}^\top\| \leq 2(1 + \sqrt{\frac{M}{N}})^2 \right\}.$$

We then have

$$\begin{aligned}
 \|\mathbb{E}\mathbf{F}_2\| &\leq \mathbb{E}\|\mathbf{F}_2\| \leq \frac{1}{N} \frac{\|\mathbf{A}\|}{d^2} \mathbb{E}\|\mathbf{X}\mathbf{B}\mathbf{X}^\top\| \\
 &\leq \frac{\|\mathbf{A}\|\|\Sigma\|}{d^2} [4(1 + \sqrt{\phi})^2 \mathbb{E}\|\mathbf{B}\|\delta(\mathcal{E}) + \frac{1}{N} \mathbb{E}\|\mathbf{B}\|\|\mathbf{Z}\mathbf{Z}^\top\|\delta(\mathcal{E}^C)] \\
 &\leq C \mathbb{E} \max_k |\tilde{b} - b_k| + \frac{1}{N} \sqrt{\mathbb{E} \max_k |\tilde{b} - b_k|^2 \mathbb{E}\|\mathbf{Z}\mathbf{Z}^\top\|^2 \delta(\mathcal{E}^C)}.
 \end{aligned}$$

By using the inequality that (see e.g. (Bai & Silverstein, 2010))

$$\mathbb{P}(\mathcal{E}^C) \leq N^{-\ell} \text{ for any } \ell > 0,$$

we have

$$\begin{aligned}
 \mathbb{E}\|\mathbf{Z}\mathbf{Z}^\top\|^2 \delta(\mathcal{E}^C) &\leq \mathbb{E}\|\mathbf{Z}\mathbf{Z}^\top\|_{\mathbb{F}}^2 \delta(\mathcal{E}^C) \leq \sqrt{\mathbb{E}\|\mathbf{Z}\mathbf{Z}^\top\|_{\mathbb{F}}^4 \mathbb{P}(\mathcal{E}^C)} \\
 &\leq N^{100} o(N^{-101}) = o(N^{-1}).
 \end{aligned} \tag{48}$$

It can be shown by Lemma B.1 that for $\ell \geq 1$,

$$\begin{aligned}
 \mathbb{P}(|\tilde{b} - b_k| > t) &\leq \frac{\mathbb{E}|\mathbf{z}_k^\top \Sigma^{1/2} \mathbf{Q}_{-k} \Sigma^{1/2} \mathbf{z}_k - \text{Tr} \Sigma \mathbf{Q}_{-k}(z)|^\ell}{(Nt)^\ell} \\
 &= t^{-\ell} \frac{\mathbb{E}[\mathbb{E}_{-k} |\mathbf{z}_k^\top \Sigma^{1/2} \mathbf{Q}_{-k} \Sigma^{1/2} \mathbf{z}_k - \text{Tr} \Sigma \mathbf{Q}_{-k}(z)|^\ell]}{N^\ell} \\
 &\leq t^{-\ell} C \frac{\mathbb{E}[(\text{Tr} \mathbf{Q}_{-k}^2)^{\ell/2} + \text{Tr}(\mathbf{Q}_{-k})^\ell]}{N^\ell} \\
 &\leq C t^{-\ell} N^{-\ell/2},
 \end{aligned}$$

where we use the fact that

$$\text{Tr}(\mathbf{Q}_{-k}(z_2))^\ell \leq M \|\mathbf{Q}_{-k}(z_2)\|^\ell \leq \frac{M}{d^\ell}.$$

By taking a large enough ℓ , we have

$$\begin{aligned}
 \mathbb{E} \max_k |\tilde{b} - b_k| &= \left(\int_{t \leq N^{-1/4}} + \int_{t > N^{-1/4}} \right) \mathbb{P}(\max_k |\tilde{b} - b_k| > t) dt \\
 &\leq N^{-1/4} + \int_{t > N^{-1/4}} \sum_{k=1}^N \mathbb{P}(|\tilde{b} - b_k| > t) dt \\
 &\leq 2N^{-1/4}.
 \end{aligned} \tag{49}$$

Similarly, one may obtain

$$\mathbb{E} \max_k |\tilde{b} - b_k|^2 = o(1). \tag{50}$$

This, along with (48) and (49), implies that

$$\|\mathbb{E}\mathbf{F}_2\| = o(1).$$

By using (41), we rewrite

$$\begin{aligned}
 \mathbb{E}\mathbf{F}_1 &= \frac{1}{N} \sum_{k=1}^N \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top \\
 &= \frac{1}{N} \sum_{k=1}^N \mathbb{E} \left[\mathbf{Q}_{-k}(z_1) - \frac{1}{N} \frac{\mathbf{Q}_{-k}(z_1) \mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-k}(z_1)}{1 + \mathbf{b}_k} \right] \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top \\
 &= \frac{1}{N} \sum_{k=1}^N \mathbb{E} \mathbf{Q}_{-k}(z_1) \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top - \frac{1}{N} \sum_{k=1}^N \mathbb{E} \frac{\frac{1}{N} \mathbf{Q}_{-k}(z_1) \mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-k}(z_1)}{1 + \mathbf{b}_k} \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top \\
 &= \mathbb{E} \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \Sigma - \frac{1}{(1 + \tilde{\mathbf{b}})N} \sum_{k=1}^N \mathbb{E} \frac{1}{N} \mathbf{Q}_{-k}(z_1) \mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-k}(z_1) \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top \\
 &\quad - \frac{1}{(1 + \tilde{\mathbf{b}})N^2} \sum_{k=1}^N \mathbb{E} \mathbf{Q}_{-k}(z_1) \mathbf{x}_k \mathbf{x}_k^\top \mathbf{Q}_{-k}(z_1) \mathbf{A} \mathbf{Q}_{-k}(z_2) \mathbf{x}_k \mathbf{x}_k^\top \frac{(\mathbf{b}_k - \tilde{\mathbf{b}})}{1 + \mathbf{b}_k} \\
 &= \mathbb{E} \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \Sigma - \frac{1}{1 + \tilde{\mathbf{b}}} (\mathbb{E}\mathbf{F}_1 + \mathbb{E}\mathbf{F}_2),
 \end{aligned} \tag{51}$$

where

$$\mathbf{F}_1 = \frac{1}{N} \mathbf{Q}_{-1}(z_1) \mathbf{x}_1 \mathbf{x}_1^\top \mathbf{Q}_{-1}(z_2) \mathbf{A} \mathbf{Q}_{-1}(z_2) \mathbf{x}_1 \mathbf{x}_1^\top,$$

$$\mathbf{F}_2 = \frac{1}{N} \mathbf{Q}_{-1}(z_1) \mathbf{x}_1 \mathbf{x}_1^\top \mathbf{Q}_{-1}(z_2) \mathbf{A} \mathbf{Q}_{-1}(z_2) \mathbf{x}_1 \mathbf{x}_1^\top \frac{(\mathbf{b}_1 - \tilde{\mathbf{b}})}{1 + \mathbf{b}_1}.$$

We first consider $\mathbb{E}\mathbf{F}_2$. Let $\tilde{\mathbf{u}}, \tilde{\mathbf{v}}$ denote a pair of unit vectors satisfying

$$\tilde{\mathbf{u}}, \tilde{\mathbf{v}} = \arg \max_{\|\tilde{\mathbf{u}}\|=\|\tilde{\mathbf{v}}\|=1} |\tilde{\mathbf{u}}^\top \mathbb{E}\mathbf{F}_2 \tilde{\mathbf{v}}|,$$

and let $\mathbf{y} = \mathbf{Q}_{-1}(z_1) \tilde{\mathbf{u}} = (y_1, \dots, y_M)^\top$. Using the Burkholder's inequality (Burkholder, 1973), we have

$$\begin{aligned}
 \mathbb{E} |\mathbf{y}^\top \mathbf{x}_1|^4 &= \mathbb{E} \left| \sum_{i=1}^M y_i x_{i1} \right|^4 \leq c \mathbb{E} \left| \sum_{i=1}^M y_i^2 \right|^2 + c \mathbb{E} \sum_{i=1}^M |y_i x_{i1}|^4 \\
 &\leq C \mathbb{E} \|\mathbf{y}\|^4 + C \mathbb{E} \sum_{i=1}^M y_i^4 \lesssim 1,
 \end{aligned}$$

where we use the inequality

$$\sum_{i=1}^M y_i^4 \leq \left(\sum_{i=1}^M y_i^2 \right)^2 \leq \|\mathbf{y}\|^4.$$

Likewise, we have $\mathbb{E} |\mathbf{x}_1^\top \tilde{\mathbf{v}}|^4 \lesssim 1$. It follows from Lemma B.1 that

$$\mathbb{E} |\mathbf{b}_1 - \tilde{\mathbf{b}}|^\ell \leq \frac{c}{N^\ell} [(\text{Tr} \mathbf{Q}_{-1}^2(z_1))^{\ell/2} + \text{Tr} \mathbf{Q}_{-1}^\ell(z_1)] \leq \frac{C}{N^{\ell/2}},$$

and

$$\mathbb{E} |\mathbf{x}_1^\top \mathbf{Q}_{-1}(z_1) \mathbf{x}_1|^\ell \leq C \mathbb{E} |\mathbf{x}_1^\top \mathbf{Q}_{-1}(z_1) \mathbf{x}_1 - \mathbb{E} \text{Tr} \Sigma \mathbf{Q}_{-1}(z_1)|^\ell + C |\mathbb{E} \text{Tr} \Sigma \mathbf{Q}_{-1}(z_1)|^\ell \lesssim N^\ell.$$

Since $b_k > 1$, we can bound the spectral norm of $\mathbb{E}F_2$ as

$$\begin{aligned}
 \|\mathbb{E}F_2\| &= |\tilde{\mathbf{u}}^\top \mathbb{E}F_2 \tilde{\mathbf{v}}| \leq \mathbb{E}|\tilde{\mathbf{u}}^\top F_2 \tilde{\mathbf{v}}| \\
 &\leq \frac{1}{N} \mathbb{E}|\tilde{\mathbf{u}}^\top \mathbf{Q}_{-1}(z_1) \mathbf{x}_1 \mathbf{x}_1^\top \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \mathbf{x}_1 \mathbf{x}_1^\top \tilde{\mathbf{v}}| |b_1 - \tilde{b}| \\
 &\leq \frac{1}{N} \sqrt{\mathbb{E}|\mathbf{y}^\top \mathbf{x}_1 \mathbf{x}_1^\top \tilde{\mathbf{v}}|^2 \mathbb{E}|\mathbf{x}_1^\top \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \mathbf{x}_1 (b_1 - \tilde{b})|^2} \\
 &\leq \frac{1}{N} \sqrt{\sqrt{\mathbb{E}|\mathbf{y}^\top \mathbf{x}_1|^4 \mathbb{E}|\mathbf{x}_1^\top \tilde{\mathbf{v}}|^4} \sqrt{\mathbb{E}|\mathbf{x}_1^\top \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \mathbf{x}_1|^4 \mathbb{E}|b_1 - \tilde{b}|^4}} \\
 &\leq C \frac{1}{N} o(N) = o(1).
 \end{aligned}$$

Therefore, it suffices to find the deterministic equivalent of $\mathbb{E}F_1$. We recall the definition above (39) that $\mathbf{A}_M = o(1)$ if $\|\mathbf{A}_M\| = o(1)$. Let $\mathbb{E}_{-1}(\cdot) = \mathbb{E}[\cdot | \mathbf{x}_2, \dots, \mathbf{x}_N]$. We have

$$\begin{aligned}
 \mathbb{E}F_1 &= \frac{1}{N} \mathbb{E} \mathbf{Q}_{-1}(z_1) \mathbf{x}_1 \mathbf{x}_1^\top \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \mathbf{x}_1 \mathbf{x}_1^\top \\
 &= \frac{1}{N} \mathbb{E} \mathbf{Q}_{-1}(z_1) [\mathbb{E}_{-1} \mathbf{x}_1 \mathbf{x}_1^\top \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \mathbf{x}_1 \mathbf{x}_1^\top] \\
 &= \frac{1}{N} \mathbb{E} \mathbf{Q}_{-1}(z_1) \Sigma^{1/2} \mathbb{E}_{-1} [\mathbf{z}_1 \mathbf{z}_1^\top \Sigma^{1/2} \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \Sigma^{1/2} \mathbf{z}_1 \mathbf{z}_1^\top] \Sigma^{1/2} \\
 &= \frac{1}{N} \mathbb{E} \mathbf{Q}_{-1}(z_1) [\text{Tr} \Sigma \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2)] \Sigma \\
 &\quad + \frac{1}{N} \mathbb{E} \mathbf{Q}_{-1}(z_1) \Sigma [\mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) + \mathbf{Q}_{-1}(z_2) \mathbf{A} \mathbf{Q}_{-1}(z_1)] \Sigma \\
 &\quad + \frac{1}{N} (\mathbb{E} z_{11}^4 - 3) \mathbb{E} \mathbf{Q}_{-1}(z_1) \Sigma^{1/2} \text{diag}(\Sigma^{1/2} \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \Sigma^{1/2}) \Sigma^{1/2} \\
 &= \frac{1}{N} \mathbb{E} [\text{Tr} \Sigma \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2)] \mathbf{Q}_{-1}(z_1) \Sigma + o(1) \\
 &= \frac{1}{N} \left[\mathbb{E} \text{Tr} \Sigma \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \right] \Pi(z_1) \Sigma + o(1),
 \end{aligned} \tag{52}$$

where the last identity is due to (42), (44) and

$$\begin{aligned}
 &\frac{1}{N} \mathbb{E} [\text{Tr} \Sigma \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2)] \mathbf{Q}_{-1}(z_1) \Sigma \\
 &= \frac{1}{N} \mathbb{E} [\text{Tr} \Sigma \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) - \mathbb{E} \text{Tr} \Sigma \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2)] \mathbf{Q}_{-1}(z_1) \Sigma \\
 &\quad + \frac{1}{N} [\mathbb{E} \text{Tr} \Sigma \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2)] \mathbf{Q}_{-1}(z_1) \Sigma \\
 &= \frac{1}{N} \mathbb{E} [\text{Tr} \Sigma \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2)] \Pi(z_1) \Sigma + o(1).
 \end{aligned}$$

By (45), (47), (51) and (52) and the fact that $\|\Pi(z_2)\|$ is bounded, we have

$$\begin{aligned}
 &\frac{1}{N} \mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \mathbf{X} \mathbf{X}^\top \Pi(z_2) \\
 &= \frac{1}{(1 + \tilde{b})} \mathbb{E} F_1 \Pi(z_2) + o(1) \\
 &= \frac{1}{1 + \tilde{b}} \left[\mathbb{E} \mathbf{Q}_{-1}(z_1) \mathbf{A} \mathbf{Q}_{-1}(z_2) \Sigma \Pi(z_2) - \frac{1}{1 + \tilde{b}} \mathbb{E} F_1 \Pi(z_2) \right] + o(1) \\
 &= \frac{\mathbb{E} \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2) \Sigma \Pi(z_2)}{1 + \frac{1}{N} \text{Tr} \Sigma \Pi(z_2)} - \frac{\frac{1}{N} [\mathbb{E} \text{Tr} \Sigma \mathbf{Q}(z_1) \mathbf{A} \mathbf{Q}(z_2)] \Pi(z_1) \Sigma \Pi(z_2)}{(1 + \frac{1}{N} \text{Tr} \Sigma \Pi(z_2))(1 + \frac{1}{N} \text{Tr} \Sigma \Pi(z_1))} + o(1).
 \end{aligned}$$

This, along with (39), (46), leads to

$$\begin{aligned} & \mathbb{E}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2) \\ &= \mathbf{\Pi}(z_1)\mathbf{A}\mathbf{\Pi}(z_2) + \frac{1}{N} \frac{[\mathbb{E}\text{Tr}\mathbf{\Sigma}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2)]\mathbf{\Pi}(z_1)\mathbf{\Sigma}\mathbf{\Pi}(z_2)}{(1 + \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_2))(1 + \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1))} + o(1). \end{aligned} \quad (53)$$

Multiplying both sides of the above equation on the left by $\mathbf{\Sigma}$, and taking the trace, we obtain

$$\begin{aligned} & \frac{1}{N} \mathbb{E}\text{Tr}\mathbf{\Sigma}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2) \\ &= \frac{1}{N} \text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1)\mathbf{A}\mathbf{\Pi}(z_2) + \frac{1}{N} \frac{[\mathbb{E}\text{Tr}\mathbf{\Sigma}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2)]\frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1)\mathbf{\Sigma}\mathbf{\Pi}(z_2)}{(1 + \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_2))(1 + \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1))} + o(1). \end{aligned}$$

It follows that

$$\begin{aligned} & \frac{1}{N} \mathbb{E}\text{Tr}\mathbf{\Sigma}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2) \\ &= \left(1 - \frac{\frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1)\mathbf{\Sigma}\mathbf{\Pi}(z_2)}{(1 + \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_2))(1 + \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1))}\right)^{-1} \frac{1}{N} \text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1)\mathbf{A}\mathbf{\Pi}(z_2) + o(1). \end{aligned} \quad (54)$$

Plugging (54) into (53), we get

$$\begin{aligned} & \mathbb{E}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2) = \mathbf{\Pi}(z_1)\mathbf{A}\mathbf{\Pi}(z_2) \\ &+ \frac{\frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1)\mathbf{A}\mathbf{\Pi}(z_2)}{(1 + \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_2))(1 + \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1)) - \frac{1}{N}\text{Tr}\mathbf{\Sigma}\mathbf{\Pi}(z_1)\mathbf{\Sigma}\mathbf{\Pi}(z_2)} \mathbf{\Pi}(z_1)\mathbf{\Sigma}\mathbf{\Pi}(z_2) + o(1). \end{aligned} \quad (55)$$

The result (15) follows by combining the (55) with (38). Now we prove (17). Using a proof analogous to that of (38), we can obtain that

$$\frac{1}{M} \text{Tr}\mathbf{C}[\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2) - \mathbb{E}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2)] = o_{a.s.}(1). \quad (56)$$

We denote the spectral decomposition of $\mathbf{C}$ by

$$\mathbf{C} = \sum_{i=1}^M \lambda_i \mathbf{u}_i \mathbf{v}_i^T.$$

By (55), we have

$$\begin{aligned} \frac{1}{M} \text{Tr}\mathbf{C}\mathbb{E}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2) &= \frac{1}{M} \text{Tr} \sum_{i=1}^M \lambda_i \mathbf{u}_i \mathbf{v}_i^T \mathbb{E}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2) \\ &= \frac{1}{M} \sum_{i=1}^M \lambda_i \mathbf{v}_i^T \mathbb{E}\mathbf{Q}(z_1)\mathbf{A}\mathbf{Q}(z_2) \mathbf{u}_i \\ &= \frac{1}{M} \sum_{i=1}^M \lambda_i \mathbf{v}_i^T \mathbf{\Pi}(z_1) \mathcal{S}(\mathbf{A}) \mathbf{\Pi}(z_2) \mathbf{u}_i + o(1) \\ &= \frac{1}{M} \text{Tr}\mathbf{C}\mathbf{\Pi}(z_1) \mathcal{S}(\mathbf{A}) \mathbf{\Pi}(z_2) + o(1). \end{aligned}$$

This, along with (56), establishes (17).

B.10. Details of Section 5.2

In Case 1 of Section 5.2, where $\mathbf{X}_1 = \alpha \mathbf{X}_2 + \tilde{\mathbf{X}}_1$, the following result holds.

Proposition B.12. *Suppose that $\tilde{\mathbf{X}}_1, \mathbf{X}_2, \varepsilon_1$ and ε_2 satisfy Assumptions 2.1-2.2. Then Theorem 3.1 continues to hold. Moreover, if we additionally impose Assumption 3.2, then Theorem 3.3 remains valid.*

Proof: We recall that $z_1 = -\lambda_t, z_2 = -\lambda_s$. We only consider h_{55} in (19) here and the remaining terms can be handled analogously. By (21), it suffices to estimate

$$\frac{1}{N} \text{Tr}(\mathbf{I}_M + z_2 \mathbf{Q}_2) \mathbf{\Sigma}_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) (\mathbf{Q}_1 + z_1 \mathbf{Q}_1^2).$$

Since

$$\mathbf{Q}_1 = \left(\frac{1}{N} \mathbf{X}_1 \mathbf{X}_1^\top - z_1 \mathbf{I}_M \right)^{-1} = \left(\frac{1}{N} \tilde{\mathbf{X}}_1 \tilde{\mathbf{X}}_1^\top - z_1 \mathbf{I}_M + \Delta \right)^{-1}, \quad \Delta = \frac{\alpha}{N} (\tilde{\mathbf{X}}_1 \mathbf{X}_2^\top + \mathbf{X}_2 \tilde{\mathbf{X}}_1^\top + \alpha \mathbf{X}_2 \mathbf{X}_2^\top).$$

We denote $\tilde{\mathbf{Q}}_1 = \left(\frac{1}{N} \tilde{\mathbf{X}}_1 \tilde{\mathbf{X}}_1^\top - z_1 \mathbf{I}_M \right)^{-1}$. Applying (43) and Lemma B.3, we have with high probability,

$$\|\mathbf{E}\| = \|\tilde{\mathbf{Q}}_1 - \mathbf{Q}_1\| = \|\tilde{\mathbf{Q}}_1 \Delta \mathbf{Q}_1\| \leq \frac{1}{|z_1|^2} \|\Delta\| \lesssim \alpha = o(1). \quad (57)$$

Then we obtain

$$\begin{aligned} \frac{z_2}{N} \text{Tr} \mathbf{Q}_2 \Sigma_2 \mathbf{Q}_1^2 &= \frac{z_2}{N} \text{Tr} \mathbf{Q}_2 \Sigma_2 (\tilde{\mathbf{Q}}_1^2 + \underbrace{\mathbf{E}^2 + \mathbf{E} \tilde{\mathbf{Q}}_1 + \tilde{\mathbf{Q}}_1 \mathbf{E}}_{\hat{\mathbf{E}}}) \\ &= \frac{z_2}{N} \text{Tr} \mathbf{Q}_2 \Sigma_2 \tilde{\mathbf{Q}}_1^2 + o_{a.s.}(1), \end{aligned}$$

where we use the fact that

$$\frac{1}{N} \text{Tr} \mathbf{Q}_2 \Sigma_2 \hat{\mathbf{E}} \leq \frac{M}{N} \|\mathbf{Q}_2 \Sigma_2 \hat{\mathbf{E}}\| \lesssim \|\hat{\mathbf{E}}\| = o_{a.s.}(1).$$

By similar argument, we have

$$h_{55} = \frac{\xi^2 \sigma^2}{N^2} \text{Tr}(\mathbf{I}_M + z_2 \mathbf{Q}_2) \Sigma_2 (\mathbf{I}_M + z_2 \mathbf{Q}_2) (\tilde{\mathbf{Q}}_1 + z_1 \tilde{\mathbf{Q}}_1^2) + o_{a.s.}(1).$$

The proof is completed.

For Case 2 in Section 5.2, we have the following proposition, which also covers the setting of self-distillation.

Proposition B.13. Suppose $\mathbf{X}_1 = \mathbf{X}_2 + \mathbf{A}$ is a signal-plus-noise data matrix, with $\|\mathbf{A}\| = o(\sqrt{N})$. The regression parameter vector $\beta = \beta_1 = \beta_2$ satisfies Assumption 3.2. When $\lambda_s \neq \lambda_t$, we have

$$\mathbf{E}\mathbf{R}(\beta_s) = \widehat{\mathbf{Bias}} + \widehat{\mathbf{Var}} + o_{a.s.}(1),$$

with

$$\widehat{\mathbf{Bias}} = \frac{a}{M} \text{Tr} \Sigma_2 [\Pi_2(-\lambda_t) - \Pi_2(-\lambda_s)] + \frac{b}{M} \text{Tr} \Sigma_2 \Pi_2'(-\lambda_t) + \frac{c}{M} \text{Tr} \Sigma_2 \Pi_2'(-\lambda_s),$$

and

$$\begin{aligned} \widehat{\mathbf{Var}} &= \frac{\xi^2 \sigma^2}{N_1} \left(d \text{Tr} \Sigma_2 [\Pi_2(-\lambda_t) - \Pi_2(-\lambda_s)] + e \text{Tr} \Sigma_2 \Pi_2'(-\lambda_t) + f \text{Tr} \Sigma_2 \Pi_2'(-\lambda_s) \right) \\ &\quad + (1 - \xi)^2 \sigma^2 \frac{1}{N_2} \text{Tr} \Sigma_2 [\Pi_2(-\lambda_s) - \lambda_s \Pi_2'(-\lambda_s)], \end{aligned}$$

where

$$\begin{aligned} a &= \frac{2\xi\lambda_1\lambda_2}{\lambda_s - \lambda_t} + \frac{2\xi\lambda_t\lambda_s(\xi\lambda_t - \lambda_s)}{(\lambda_s - \lambda_t)^2} - \frac{2\xi\lambda_s^2\lambda_t^2}{(\lambda_s - \lambda_t)^3}, \quad b = \xi^2\lambda_t^2 - \frac{2\xi^2\lambda_t^2\lambda_s}{\lambda_t - \lambda_s}, \\ c &= \lambda_s^2 - \frac{2\xi\lambda_t\lambda_s^2}{\lambda_t - \lambda_s}, \quad d = \frac{2\lambda_s}{\lambda_t - \lambda_s} + \frac{\lambda_s^2}{(\lambda_t - \lambda_s)^2} + \frac{\lambda_t\lambda_s^2}{(\lambda_s - \lambda_t)^3} + \frac{2\lambda_s\lambda_t}{(\lambda_t - \lambda_s)^2}, \\ e &= -\lambda_t + \frac{2\lambda_s\lambda_t}{(\lambda_t - \lambda_s)^2} - \frac{\lambda_t\lambda_s^2}{(\lambda_s - \lambda_t)^2}, \quad f = \frac{\lambda_s^2}{\lambda_s - \lambda_t} - \frac{\lambda_t\lambda_s^2}{(\lambda_s - \lambda_t)^2}. \end{aligned}$$

When $\lambda = \lambda_s = \lambda_t$, $\widehat{\mathbf{Bias}}$ is given in (59) and

$$\widehat{\mathbf{Var}} = (1 - \xi)^2 \sigma^2 \frac{1}{N_2} \text{Tr} \Sigma_2 (\Pi_2 + \lambda \Pi_2') + \frac{\xi^2 \sigma^2}{N_1} \text{Tr} \Sigma_2 [\Pi_2 - 3\lambda \Pi_2' + 3\lambda^2 \Pi_2^{(2)} - \lambda^3 \Pi_2^{(3)}],$$

with $\Pi_2^{(k)} = \left. \frac{d^k \Pi_2(z)}{dz^k} \right|_{z=-\lambda}$.

Proof: By an argument analogous to that used for (57), one may readily verify that

$$\|\mathbf{Q}_1(z) - \mathbf{Q}_2(z)\| = o(1).$$

Then (26) becomes

$$\begin{aligned} \frac{1}{M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{H} \mathbf{H}^\top &= \frac{1}{M} \left[\xi^2 z_1^2 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2^2(z_1) + 2\xi z_1 z_2 \underbrace{\text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2(z_1) \mathbf{Q}_2(z_2)}_{t_1} + 2\xi^2 z_1^2 z_2 \underbrace{\text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2(z_2) \mathbf{Q}_2^2(z_1)}_{t_2} \right. \\ &\quad + z_2^2 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2^2(z_2) + 2\xi z_1 z_2^2 \underbrace{\text{Tr} [\mathbf{Q}_2(z_2) \mathbf{\Sigma}_2 \mathbf{Q}_2(z_2) \mathbf{Q}_2(z_1)]}_{t_3} \\ &\quad \left. + \xi^2 z_1^2 z_2^2 \underbrace{\text{Tr} [\mathbf{Q}_2(z_2) \mathbf{\Sigma}_2 \mathbf{Q}_2(z_2) \mathbf{Q}_2^2(z_1)]}_{t_4} \right] + o(1). \end{aligned}$$

We note that when $z_1 \neq z_2$,

$$\mathbf{Q}_2(z_1) - \mathbf{Q}_2(z_2) = (z_1 - z_2) \mathbf{Q}_2(z_1) \mathbf{Q}(z_2), \quad \mathbf{Q}_2(z_1) \mathbf{Q}_2(z_2) = \mathbf{Q}_2(z_2) \mathbf{Q}_2(z_1).$$

Then we have

$$\begin{aligned} \frac{1}{M} t_1 &= \frac{2\xi z_1 z_2}{M} \text{Tr} \mathbf{\Sigma}_2 \frac{\mathbf{Q}_2(z_1) - \mathbf{Q}_2(z_2)}{z_1 - z_2} = \frac{2\xi z_1 z_2}{M(z_1 - z_2)} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{\Pi}_2(z_1) - \mathbf{\Pi}_2(z_2)] + o_{a.s.}(1), \\ \frac{1}{M} t_2 &= \frac{2\xi^2 z_1^2 z_2}{M(z_1 - z_2)} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{Q}_2(z_1) - \mathbf{Q}_2(z_2)] \mathbf{Q}_2(z_1) \\ &= -\frac{2\xi^2 z_1^2 z_2}{M(z_1 - z_2)} \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}'_2(z_1) - \frac{2\xi^2 z_1^2 z_2}{M(z_1 - z_2)^2} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{\Pi}_2(z_1) - \mathbf{\Pi}_2(z_2)] + o_{a.s.}(1), \\ \frac{1}{M} t_3 &= \frac{2\xi z_1 z_2^2}{M(z_1 - z_2)} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{Q}_2(z_1) - \mathbf{Q}_2(z_2)] \mathbf{Q}_2(z_2) \\ &= -\frac{2\xi z_1 z_2^2}{M(z_1 - z_2)} \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}'_2(z_2) + \frac{2\xi z_1 z_2^2}{M(z_1 - z_2)^2} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{\Pi}_2(z_1) - \mathbf{\Pi}_2(z_2)] + o_{a.s.}(1), \\ \frac{1}{M} t_4 &= \frac{\xi^2 z_1^2 z_2^2}{M(z_1 - z_2)^2} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{Q}_2(z_1) - \mathbf{Q}_2(z_2)]^2 \\ &= \frac{\xi^2 z_1^2 z_2^2}{M(z_1 - z_2)^2} \text{Tr} \mathbf{\Sigma}_2 \left[\mathbf{Q}_2^2(z_2) + \mathbf{Q}_2^2(z_1) - 2 \frac{\mathbf{Q}_2(z_1) - \mathbf{Q}_2(z_2)}{z_1 - z_2} \right] \\ &= \frac{\xi^2 z_1^2 z_2^2}{M(z_1 - z_2)^2} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{\Pi}'_2(z_1) + \mathbf{\Pi}'_2(z_2)] - \frac{2\xi^2 z_1^2 z_2^2}{M(z_1 - z_2)^3} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{\Pi}_2(z_1) - \mathbf{\Pi}_2(z_2)] + o_{a.s.}(1). \end{aligned}$$

Based on above results, we have $\frac{1}{M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{H} \mathbf{H}^\top = \widehat{\mathbf{Bias}} + o_{a.s.}(1)$. As for the variance, one may check that

$$\begin{aligned} h_{55} &= \frac{\xi^2 \sigma^2}{N_1} \text{Tr} \left[\mathbf{\Sigma}_2 [\mathbf{\Pi}_2(z_1) + z_1 \mathbf{\Pi}'_2(z_1)] + 2z_2(z_2 - z_1)^{-1} \mathbf{\Sigma}_2 [\mathbf{\Pi}_2(z_2) - \mathbf{\Pi}_2(z_1)] \right. \\ &\quad + z_2^2(z_2 - z_1)^{-1} \mathbf{\Sigma}_2 \mathbf{\Pi}'_2(z_2) + z_2^2(z_1 - z_2)^{-2} \mathbf{\Sigma}_2 [\mathbf{\Pi}_2(z_1) - \mathbf{\Pi}_2(z_2)] - 2z_1 z_2(z_2 - z_1)^{-2} \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}'_2(z_1) \\ &\quad + 2z_1 z_2(z_1 - z_2)^{-2} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{\Pi}_2(z_2) - \mathbf{\Pi}_2(z_1)] \\ &\quad \left. + z_1 z_2^2(z_1 - z_2)^{-2} \text{Tr} \mathbf{\Sigma}_2 [\mathbf{\Pi}'_2(z_1) + \mathbf{\Pi}'_2(z_2)] - z_1 z_2^2(z_1 - z_2)^3 \text{Tr} [\mathbf{\Pi}_2(z_1) - \mathbf{\Pi}_2(z_2)] \right] + o_{a.s.}(1), \end{aligned} \quad (58)$$

and the limit of h_{66} is the same as that in (24), where h_{55}, h_{66} are given in Appendix B.2.

When $\lambda = \lambda_t = \lambda_s$, denoting $\mathbf{Q}_2 = \mathbf{Q}_2(-\lambda)$, we have

$$\begin{aligned} \frac{1}{M} \text{Tr} \mathbf{\Sigma}_2 \mathbf{H} \mathbf{H}^\top &= \frac{1}{M} \left[(1 + \xi)^2 \lambda^2 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2^2 - (2\xi + \xi^2) \lambda^3 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2^3 + \xi^2 \lambda^4 \text{Tr} \mathbf{\Sigma}_2 \mathbf{Q}_2^4 \right] \\ &= \underbrace{\frac{1}{M} \left[(1 + \xi)^2 \lambda^2 \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}'_2 - (2\xi + \xi^2) \lambda^3 \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}_2^{(2)} + \xi^4 \lambda^4 \text{Tr} \mathbf{\Sigma}_2 \mathbf{\Pi}_2^{(3)} \right]}_{\mathbf{Bias}} + o_{a.s.}(1), \end{aligned} \quad (59)$$

where we use Vitali's convergence theorem. Similarly, we have

$$h_{55} = \frac{\xi^2 \sigma^2}{N_1} \text{Tr} \Sigma_2 [\Pi_2 + 3z \Pi_2' + 3z^2 \Pi_2^{(2)} + z^3 \Pi_2^{(3)}] + o_{a.s.}(1),$$

and the limit of h_{66} coincides with the one given in (24). The proof is completed.

C. Additional Experimental Details

C.1. Impact of regularization parameters

To examine the impact of the regularization parameters λ_t, λ_s , we plot the empirical excess risk of the student model for $(\lambda_t, \lambda_s) \in [0.01, 0.5]^2$ in Figures 10-12 (averaged over 5 trials), corresponding to $\xi = 0.5, -0.5$ and 1.5 , respectively. We set $\beta_1 = \beta_2 \sim \mathcal{N}(0, \frac{1}{M} \mathbf{I}_M)$, $(M, N_1, N_2) = (400, 300, 200)$, $\sigma^2 = 1$. We consider two scenarios: (i) the case without covariate shift, where $\Sigma_2 = \Sigma_1 = \mathbf{I}_M$; and (ii) the case with covariate shift, where $\Sigma_2 = \mathbf{I}_M, \Sigma_1 = \text{diag}(d_1, \dots, d_M)$ with

$$d_i = 0.64\delta(i \leq M/2) + 0.25\delta(M/2 < i \leq M).$$

These figures reveal that the relative importance of λ_t and λ_s is highly sensitive to the value of ξ . Specifically, when $\xi > 1$, the teacher's regularization parameter λ_t exerts a dominant influence on the risk. Conversely, for $\xi = -0.5$, the risk variation is governed almost entirely by λ_s , reflecting the diminished impact of the teacher's guidance in the anti-learning regime.

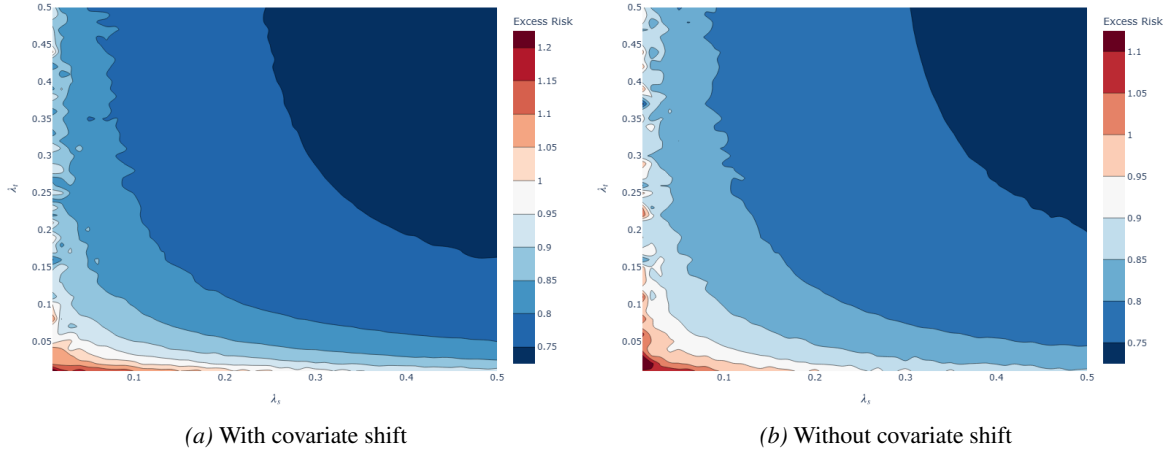


Figure 10. Excess risk when $\xi = 0.5$.

C.2. Nonlinear models

When considering more complex models, we fix the student model and let the teacher model be a deeper fully connected neural network:

$$f_{\text{NN}}^t = \mathbf{a}_t^T \sigma(\widetilde{\mathbf{W}}_3 \sigma(\widetilde{\mathbf{W}}_2 \sigma(\widetilde{\mathbf{W}}_1 \mathbf{x}))),$$

where

$$\mathbf{a}_t = \arg \min_{\mathbf{a}} \frac{1}{N_1} \|\mathbf{y}_1 - [\sigma(\widetilde{\mathbf{W}}_3 \sigma(\widetilde{\mathbf{W}}_2 \sigma(\widetilde{\mathbf{W}}_1 \mathbf{X}_1))]^T \mathbf{a}\|^2 + \lambda_t \|\mathbf{a}\|^2.$$

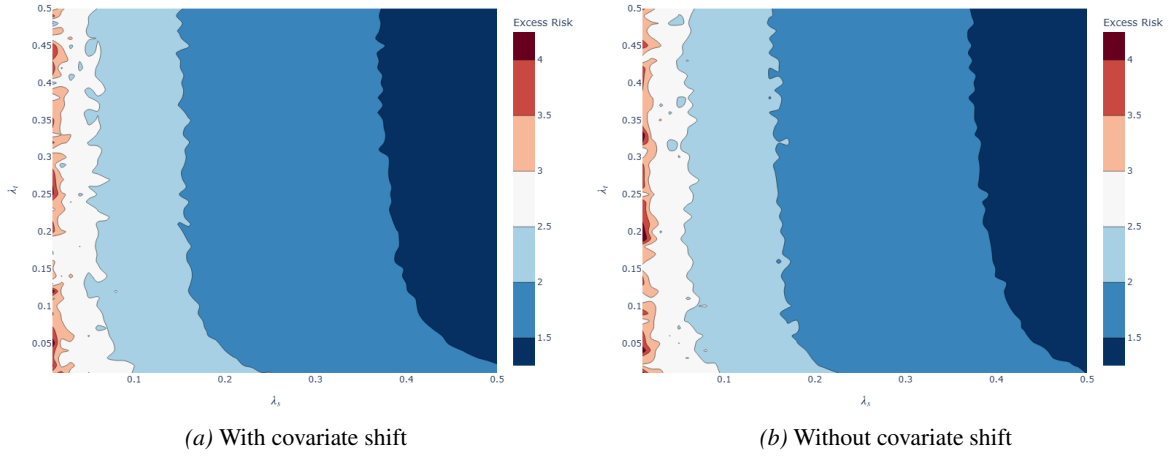


Figure 11. Excess risk when $\xi = -0.5$.

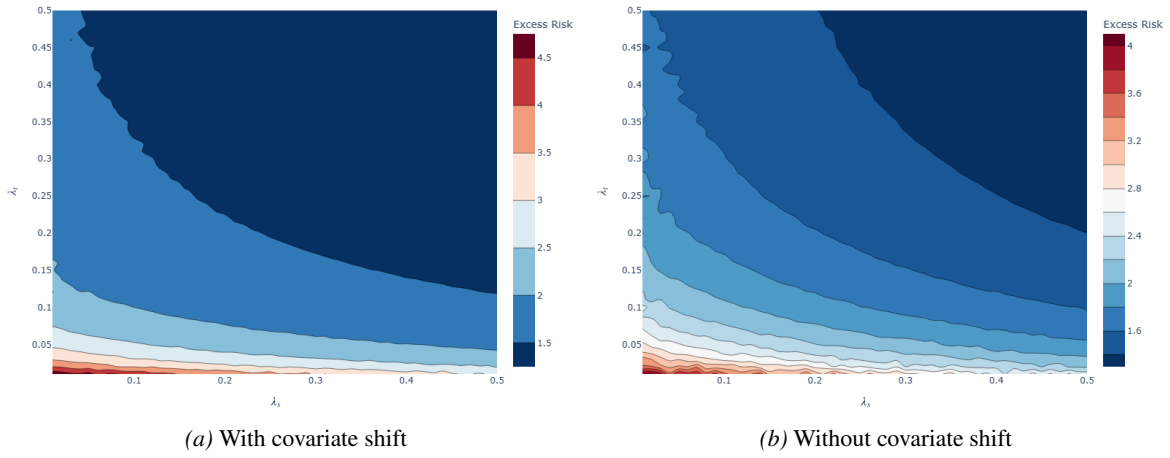


Figure 12. Excess risk when $\xi = 1.5$.

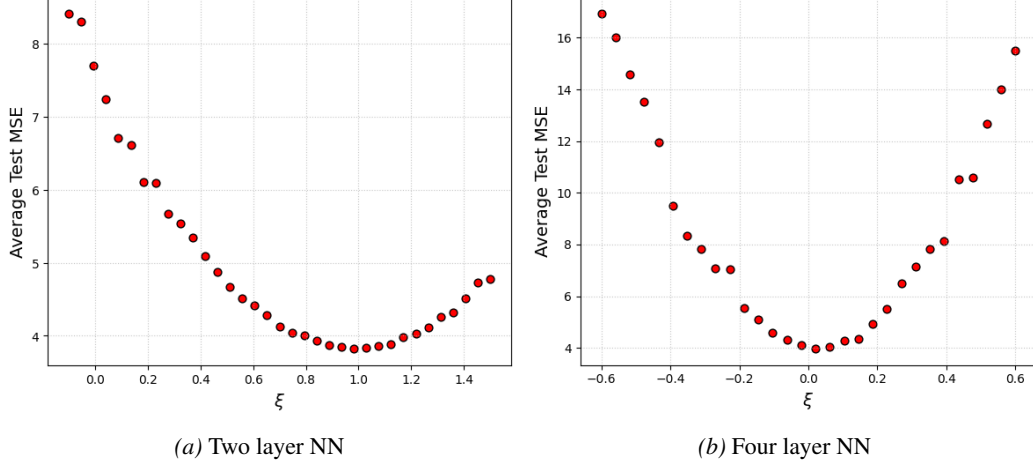


Figure 13. Excess risk estimated over 30 trials. We set $(M, N_1, N_2) = (50, 200, 100)$. (a) Settings: $(\lambda_t, \lambda_s) = (0.5, 0.2)$, $\sigma(x) = x^3$. The weight matrices $\widetilde{\mathbf{W}} \in \mathbb{R}^{n_1 \times M}$ and $\mathbf{W} \in \mathbb{R}^{n \times M}$ have i.i.d. centered Gaussian entries with variance M^{-1} , where $(n, n_1) = (100, 200)$. (b) Settings: $\lambda_t = \lambda_s = 0.2$, $(n_0, n_1, n_2, n_3) = (M, 600, 400, 200)$. The weight matrices $\widetilde{\mathbf{W}}_i \in \mathbb{R}^{n_i \times n_{i-1}}$ have i.i.d. centered Gaussian random variables with variance n_{i-1}^{-1} . We use the Leaky ReLU activation: $\sigma(x) = 0.01x\delta(x \leq 0) + x\delta(x > 0)$.

C.3. Demonstration of Corollary 3.8

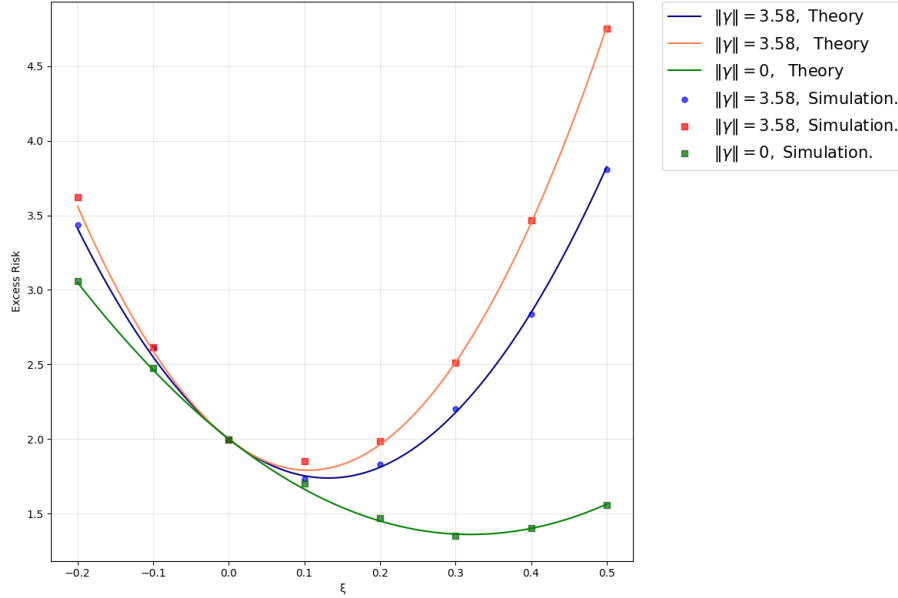


Figure 14. Theoretical predictions (solid curves) versus simulation results (scatter points, averaged over 100 independent trials) for ridgeless regression. We set $(M, N_1, N_2) = (400, 600, 600)$, $\beta_2 = \frac{4}{\sqrt{M}}(1, \dots, 1)^T$, $\sigma^2 = 1$ and $\Sigma_2 = \mathbf{I}_M$. We label the case $\|\gamma\| = 3.58$ as $\gamma = \frac{-8}{\sqrt{M}}(1, \dots, 1, 0, \dots, 0)^T$ with the first $M/5$ entries equal to 1. The orange and green curves correspond to the setting where $\Sigma_1 = \text{diag}(4, \dots, 4, \frac{1}{4}, \dots, \frac{1}{4})$, with the first half of the diagonal entries equal to 4 and the second half equal to $\frac{1}{4}$. The dark blue curve corresponds to the setting where $\Sigma_1 = 4\mathbf{I}_M$.

Figure 14 presents empirical results that support 3.8. The gap between the orange and green curves quantifies the impact of model shift on the excess risk. Furthermore, the gap between the dark blue and orange curves reflects the role of the term $\text{Tr}\Sigma_1^{-1}\Sigma_2$ as characterized in Corollary 3.8.